

Mathematics of Machine Learning

Over-Fitting



Overfitting, regularisation, feature selection

Outline

1. **Over-fitting?**
2. Controlling Complexity
3. Hidden structure
4. Regularisation



Over-fitting

- Complex machine can **over-fit**

over-fitting: fitting the training data well at the cost of getting poorer generalisation performance

- Three red cars. . .
- If we used an infinitely flexible machine we can fit our training data perfectly, but would have no generalisation ability

Over-fitting

- Complex machine can **over-fit**

***over-fitting:** fitting the training data well at the cost of getting poorer generalisation performance*

- Three red cars. . .
- If we used an infinitely flexible machine we can fit our training data perfectly, but would have no generalisation ability

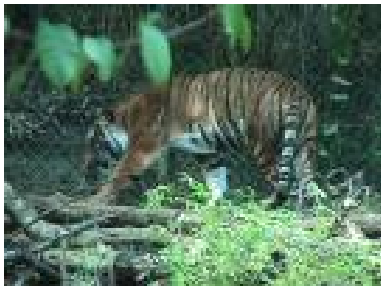
Over-fitting

- Complex machine can **over-fit**

***over-fitting:** fitting the training data well at the cost of getting poorer generalisation performance*

- Three red cars. . .
- If we used an infinitely flexible machine we can fit our training data perfectly, but would have no generalisation ability

Binary Classification Task for You



Class 1



Class 2

Which Category?

- Which category does the following image belong to?



Spurious Rules

- You ask a learning machine to solve a task based on data
- It will find a rule that does this, but not necessary the rule you had in mind
- Infinitely flexible machines have an infinity of spurious rules they can learn
- What should we do?

Spurious Rules

- You ask a learning machine to solve a task based on data
- It will find a rule that does this, but not necessary the rule you had in mind
- Infinitely flexible machines have an infinity of spurious rules they can learn
- What should we do?

Spurious Rules

- You ask a learning machine to solve a task based on data
- It will find a rule that does this, but not necessary the rule you had in mind—machine learning can't read your mind
- Infinitely flexible machines have an infinity of spurious rules they can learn
- What should we do?

Spurious Rules

- You ask a learning machine to solve a task based on data
- It will find a rule that does this, but not necessary the rule you had in mind—machine learning can't read your mind
- Infinitely flexible machines have an infinity of spurious rules they can learn
- What should we do?

Spurious Rules

- You ask a learning machine to solve a task based on data
- It will find a rule that does this, but not necessary the rule you had in mind—machine learning can't read your mind
- Infinitely flexible machines have an infinity of spurious rules they can learn—they are useless
- What should we do?

Spurious Rules

- You ask a learning machine to solve a task based on data
- It will find a rule that does this, but not necessary the rule you had in mind—machine learning can't read your mind
- Infinitely flexible machines have an infinity of spurious rules they can learn—they are useless
- What should we do?

All Binary Functions

$$\mathbf{x}_0 = 000 \quad y_0 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_1 = 100 \quad y_1 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_2 = 010 \quad y_2 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_3 = 110 \quad y_3 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_4 = 001 \quad y_4 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_5 = 101 \quad y_5 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_6 = 011 \quad y_6 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_7 = 111 \quad y_7 = \begin{cases} 0 \\ 1 \end{cases}$$

All Binary Functions

$$\mathbf{x}_0 = 000 \quad y_0 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_1 = 100 \quad y_1 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_2 = 010 \quad y_2 = \begin{cases} \textcolor{blue}{\times} \\ 1 \end{cases}$$

$$\mathbf{x}_3 = 110 \quad y_3 = \begin{cases} \textcolor{blue}{\times} \\ 1 \end{cases}$$

$$\mathbf{x}_4 = 001 \quad y_4 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_5 = 101 \quad y_5 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_6 = 011 \quad y_6 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_7 = 111 \quad y_7 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathcal{D} = \{(000, 0), (010, 1), (110, 1), (001, 0), (101, 0)\}$$

All Binary Functions

$$\mathbf{x}_0 = 000 \quad y_0 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_1 = 100 \quad y_1 = \begin{cases} 0 \\ 1 \end{cases} \quad \textcolor{red}{\text{unseen}}$$

$$\mathbf{x}_2 = 010 \quad y_2 = \begin{cases} \textcolor{blue}{\times} \\ 1 \end{cases}$$

$$\mathbf{x}_3 = 110 \quad y_3 = \begin{cases} \textcolor{blue}{\times} \\ 1 \end{cases}$$

$$\mathbf{x}_4 = 001 \quad y_4 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_5 = 101 \quad y_5 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_6 = 011 \quad y_6 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_7 = 111 \quad y_7 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathcal{D} = \{(000, 0), (010, 1), (110, 1), (001, 0), (101, 0)\}$$

All Binary Functions

$$\mathbf{x}_0 = 000 \quad y_0 = \begin{cases} 0 \\ \text{X} \end{cases}$$

$$\mathbf{x}_1 = 100 \quad y_1 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_2 = 010 \quad y_2 = \begin{cases} \text{X} \\ 1 \end{cases}$$

$$\mathbf{x}_3 = 110 \quad y_3 = \begin{cases} \text{X} \\ 1 \end{cases}$$

$$\mathbf{x}_4 = 001 \quad y_4 = \begin{cases} 0 \\ \text{X} \end{cases}$$

$$\mathbf{x}_5 = 101 \quad y_5 = \begin{cases} 0 \\ \text{X} \end{cases}$$

$$\mathbf{x}_6 = 011 \quad y_6 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_7 = 111 \quad y_7 = \begin{cases} 0 \\ 1 \end{cases} \quad \text{unseen}$$

$$\mathcal{D} = \{(000, 0), (010, 1), (110, 1), (001, 0), (101, 0)\}$$

All Binary Functions

$$\mathbf{x}_0 = 000 \quad y_0 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_1 = 100 \quad y_1 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_2 = 010 \quad y_2 = \begin{cases} \textcolor{blue}{\times} \\ 1 \end{cases}$$

$$\mathbf{x}_3 = 110 \quad y_3 = \begin{cases} \textcolor{blue}{\times} \\ 1 \end{cases}$$

$$\mathbf{x}_4 = 001 \quad y_4 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases} \quad \textcolor{red}{\text{seen}}$$

$$\mathbf{x}_5 = 101 \quad y_5 = \begin{cases} 0 \\ \textcolor{blue}{\times} \end{cases}$$

$$\mathbf{x}_6 = 011 \quad y_6 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathbf{x}_7 = 111 \quad y_7 = \begin{cases} 0 \\ 1 \end{cases}$$

$$\mathcal{D} = \{(000, 0), (010, 1), (110, 1), (001, 0), (101, 0)\}$$

Are MLPs Universal Approximators?

Are MLPs Universal Approximators?

- Yes

Are MLPs Universal Approximators?

- Yes and No

Are MLPs Universal Approximators?

- Yes and No
- Yes: If you give me any function, I can find an MLP that approximates that function to any desired accuracy

Are MLPs Universal Approximators?

- Yes and No
- Yes: If you give me any function, I can find an MLP that approximates that function to any desired accuracy
- No: If you give me an MLP, I can find a function with an arbitrary high approximation error

Are MLPs Universal Approximators?

- Yes and No
- Yes: If you give me any function, I can find an MLP that approximates that function to any desired accuracy
- No: If you give me an MLP, I can find a function with an arbitrary high approximation error
- Would an MLP that could approximate any function be useful?

Are MLPs Universal Approximators?

- Yes and No
- Yes: If you give me any function, I can find an MLP that approximates that function to any desired accuracy
- No: If you give me an MLP, I can find a function with an arbitrary high approximation error
- Would an MLP that could approximate any function be useful?
- **Absolutely not!**

Outline

1. Over-fitting?
2. **Controlling Complexity**
3. Hidden structure
4. Regularisation



Controlling Complexity

- Infinitely flexible machine don't generalise
- Machine learning only works because there is some structure in the data
- A successful machine should capture this structure
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

Controlling Complexity

- Infinitely flexible machine don't generalise (because any unseen data could have any value)
- **Machine learning only works because there is some structure in the data**
- A successful machine should capture this structure
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

Controlling Complexity

- Infinitely flexible machine don't generalise (because any unseen data could have any value)
- **Machine learning only works because there is some structure in the data**
- A successful machine should capture this structure
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

Controlling Complexity

- Infinitely flexible machine don't generalise (because any unseen data could have any value)
- **Machine learning only works because there is some structure in the data**
- **A successful machine should capture this structure**
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

Controlling Complexity

- Infinitely flexible machine don't generalise (because any unseen data could have any value)
- **Machine learning only works because there is some structure in the data**
- A successful machine should capture this structure
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

Controlling Complexity

- Infinitely flexible machine don't generalise (because any unseen data could have any value)
- **Machine learning only works because there is some structure in the data**
- A successful machine should capture this structure
- Even deep learning machines with millions or billions of parameters only work because they successfully capture the structure of images or text
- Different learning machines have different performance on different problems because the data has different structure

Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets—but this is only a part of their success
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets—but this is only a part of their success
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

Training Examples

- As we increase the number of training examples, we make it hard to find a spurious rule
- Bigger data sets allow us to use more complicated machines
- Part of the success of deep learning is because they use huge training sets—but this is only a part of their success
- (Labelled) data is often expensive to collect so we sometimes have no choice but to use a small training set
- One of the limitations of using deep learning comes because we often have limited data

Identifying Structure

- In some cases we know *a priori* some of the structure in the data
- In images we believe the identity of an object is invariant to translation and scaling
- The success of *convolutional neural networks* (CNNs) in deep learning is in large part because the convolutions respect translational invariance

Identifying Structure

- In some cases we know *a priori* some of the structure in the data
- In images we believe the identity of an object is invariant to translation and scaling
- The success of *convolutional neural networks* (CNNs) in deep learning is in large part because the convolutions respect translational invariance

Identifying Structure

- In some cases we know *a priori* some of the structure in the data
- In images we believe the identity of an object is invariant to translation and scaling
- The success of *convolutional neural networks* (CNNs) in deep learning is in large part because the convolutions respect translational invariance

Preprocessing

- Structure might often be obscure to the learning machine
- If we are trying to predict the spread of disease then a list of place names might be a lot less useful than their coordinates
- Imposing an ordering on an unordered set might **not** be useful

$\{ \text{"blue"} : 0, \text{"brown"} : 1, \text{"green"} : 2, \text{"black"} : 3 \}$

- Choosing an encoding that reflect meaningful structure is essential to successful machine learning

Preprocessing

- Structure might often be obscure to the learning machine
- If we are trying to predict the spread of disease then a list of place names might be a lot less useful than their coordinates
- Imposing an ordering on an unordered set might **not** be useful

$\{ \text{“blue”} : 0, \text{“brown”} : 1, \text{“green”} : 2, \text{“black”} : 3 \}$

- Choosing an encoding that reflect meaningful structure is essential to successful machine learning

Preprocessing

- Structure might often be obscure to the learning machine
- If we are trying to predict the spread of disease then a list of place names might be a lot less useful than their coordinates
- Imposing an ordering on an unordered set might **not** be useful

$\{ \text{"blue"} : 0, \text{"brown"} : 1, \text{"green"} : 2, \text{"black"} : 3 \}$

- Choosing an encoding that reflect meaningful structure is essential to successful machine learning

Preprocessing

- Structure might often be obscure to the learning machine
- If we are trying to predict the spread of disease then a list of place names might be a lot less useful than their coordinates
- Imposing an ordering on an unordered set might **not** be useful

$\{ \text{“blue”} : 0, \text{“brown”} : 1, \text{“green”} : 2, \text{“black”} : 3 \}$

- Choosing an encoding that reflect meaningful structure is essential to successful machine learning

Automatic Preprocessing

- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
- Having very large datasets and simple filters (the number of weights in the CNN layers tends to be small) stops overfitting

Automatic Preprocessing

- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
- Having very large datasets and simple filters (the number of weights in the CNN layers tends to be small) stops overfitting

Automatic Preprocessing

- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
- Having very large datasets and simple filters (the number of weights in the CNN layers tends to be small) stops overfitting

Automatic Preprocessing

- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised (CNN finds localised filter)
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
- Having very large datasets and simple filters (the number of weights in the CNN layers tends to be small) stops overfitting

Automatic Preprocessing

- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised (CNN finds localised filter)
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
- Having very large datasets and simple filters (the number of weights in the CNN layers tends to be small) stops overfitting

Automatic Preprocessing

- One view of deep learning is that each layer (particularly in CNNs) acts as a preprocessor
- That is, it finds filters that captures features salient to the problem being tackled
- For both images and texts we expect salient features to be spatially localised (CNN finds localised filter)
- The deep structure allows ever more complicated features to be captured—that is, we can find spatially localised features on different scales
- Having very large datasets and simple filters (the number of weights in the CNN layers tends to be small) stops overfitting

Outline

1. Over-fitting?
2. Controlling Complexity
3. **Hidden structure**
4. Regularisation



Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important

Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important

Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best (stupid but effective)
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important

Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best (stupid but effective)
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important

Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best (stupid but effective)
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important

Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best (stupid but effective)
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important

Hidden Structure

- Often the structure of data is invisible to us
- A very successful strategy is to try many different machine learning techniques and choose the best (stupid but effective)
- Often learning machines have adjustable parameters (hyper-parameters) that we have to set (they are the same for all input data, but change with the problem)
- We need to choose the hyper-parameters to fit the data in our problem
- Fine tuning hyper-parameter is important and almost always required (true in SVMs, MLP, deep learning)

Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!**
- Need to split your data up into training and validation set
- Use the validation set to choose the hyper-parameters
- You need a separate testing set if you want to measure your generalisation performance

Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!**
- Need to split your data up into training and validation set
- Use the validation set to choose the hyper-parameters
- You need a separate testing set if you want to measure your generalisation performance

Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!—you will overfit**
- Need to split your data up into training and validation set
- Use the validation set to choose the hyper-parameters
- You need a separate testing set if you want to measure your generalisation performance

Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!**—you will overfit
- Need to split your data up into training and validation set
- Use the validation set to choose the hyper-parameters
- You need a separate testing set if you want to measure your generalisation performance

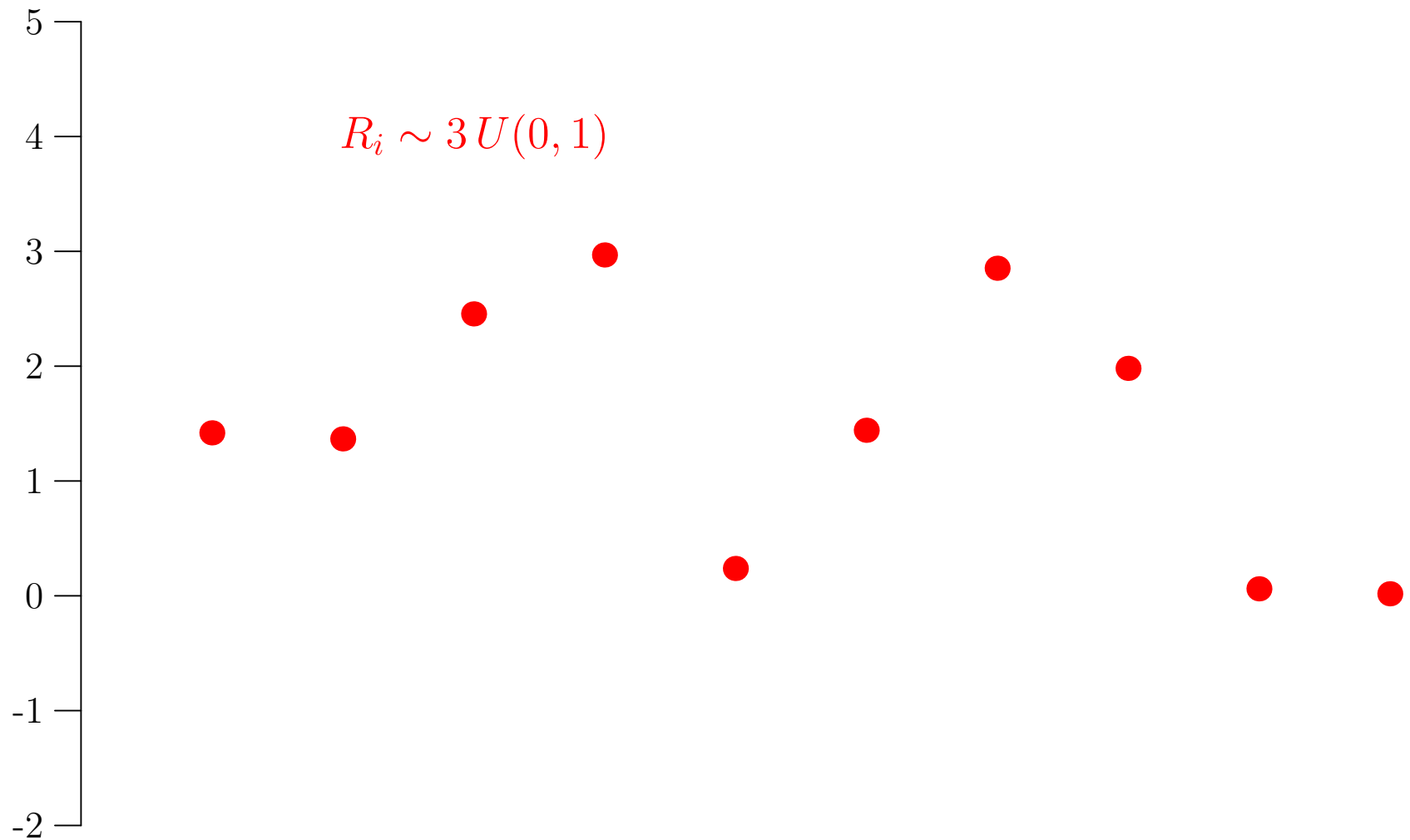
Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!**—you will overfit
- Need to split your data up into training and validation set
- Use the validation set to choose the hyper-parameters
- You need a separate testing set if you want to measure your generalisation performance

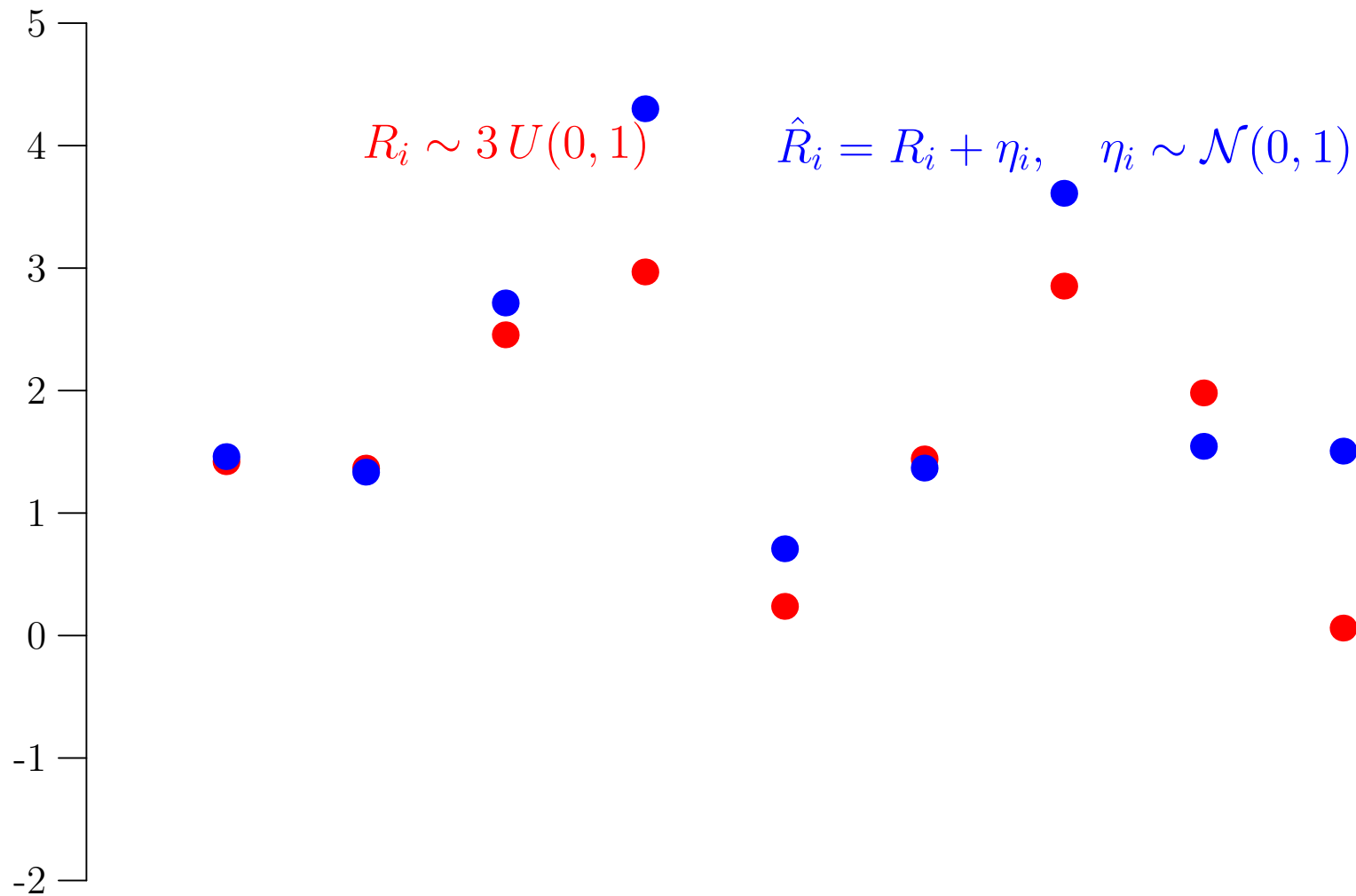
Measuring Generalisation Performance

- Recall, we want to predict **unseen** data
- **You cannot use data that you have trained on!**—you will overfit
- Need to split your data up into training and validation set
- Use the validation set to choose the hyper-parameters
- You need a separate testing set if you want to measure your generalisation performance

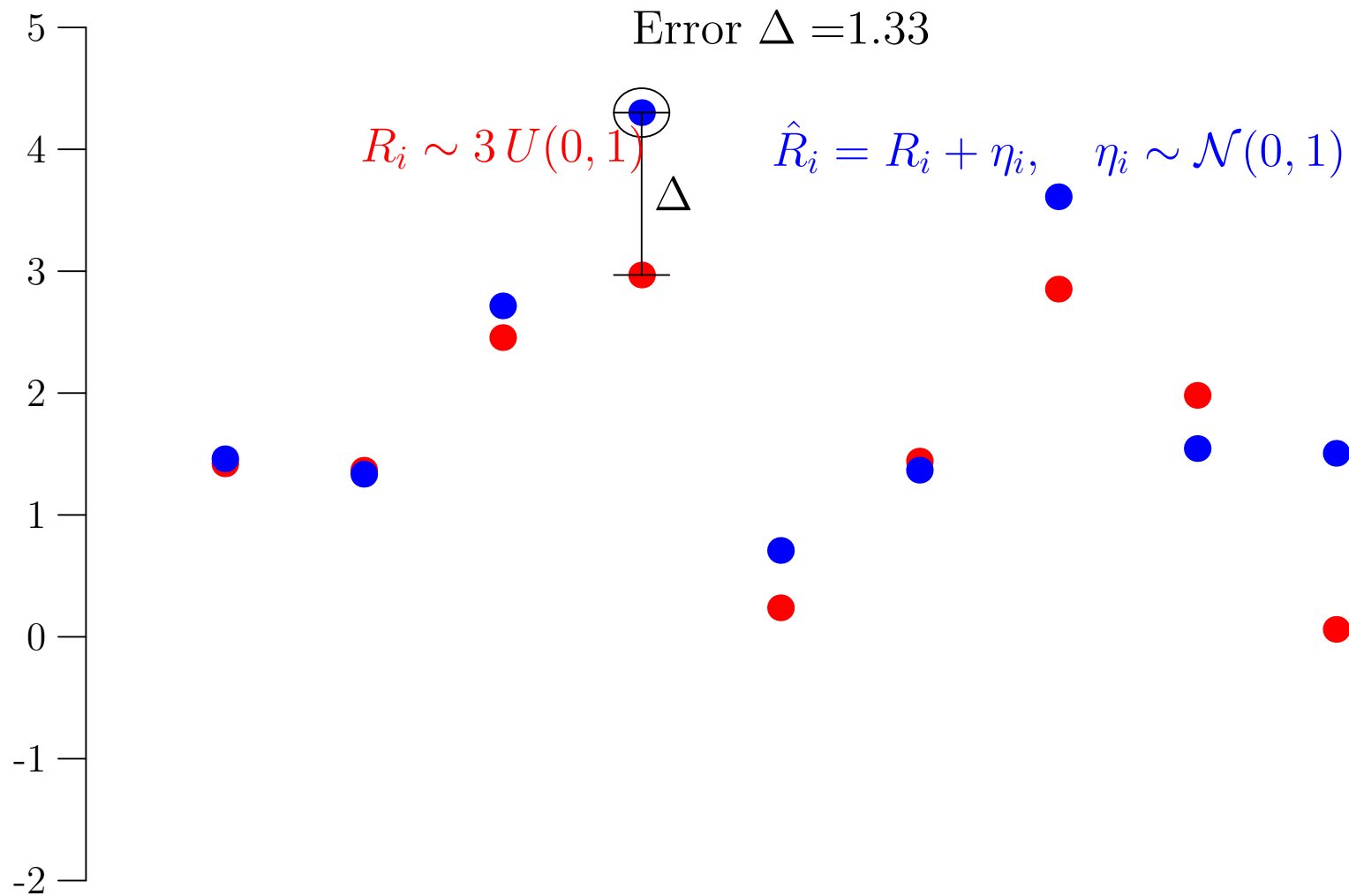
The Overfitting Game



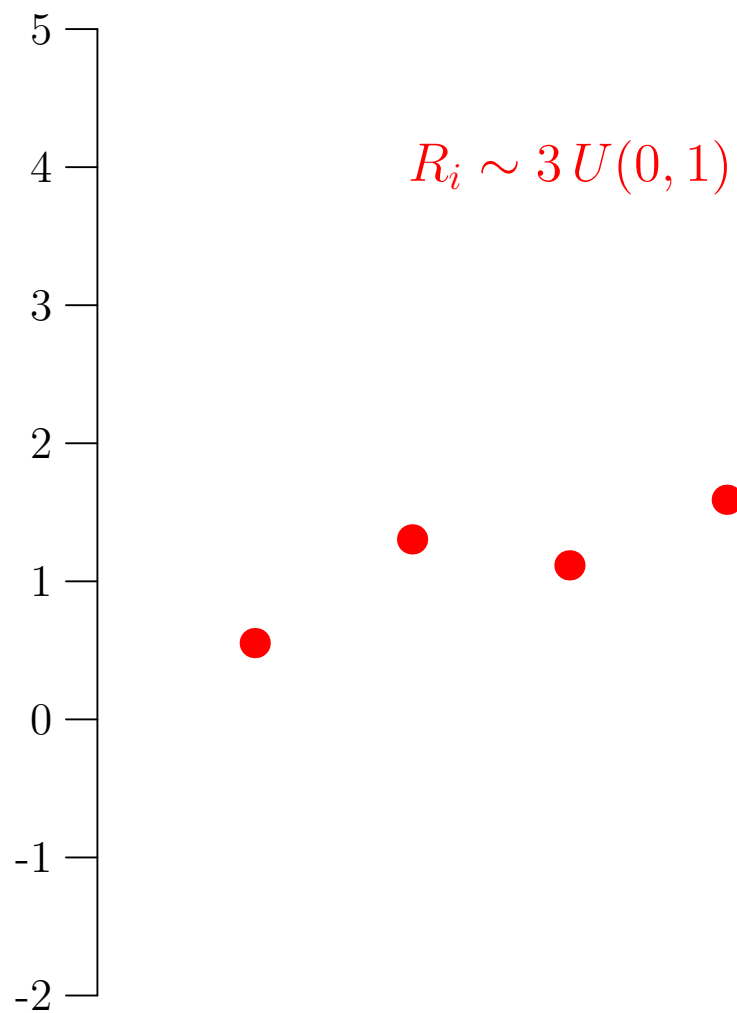
The Overfitting Game



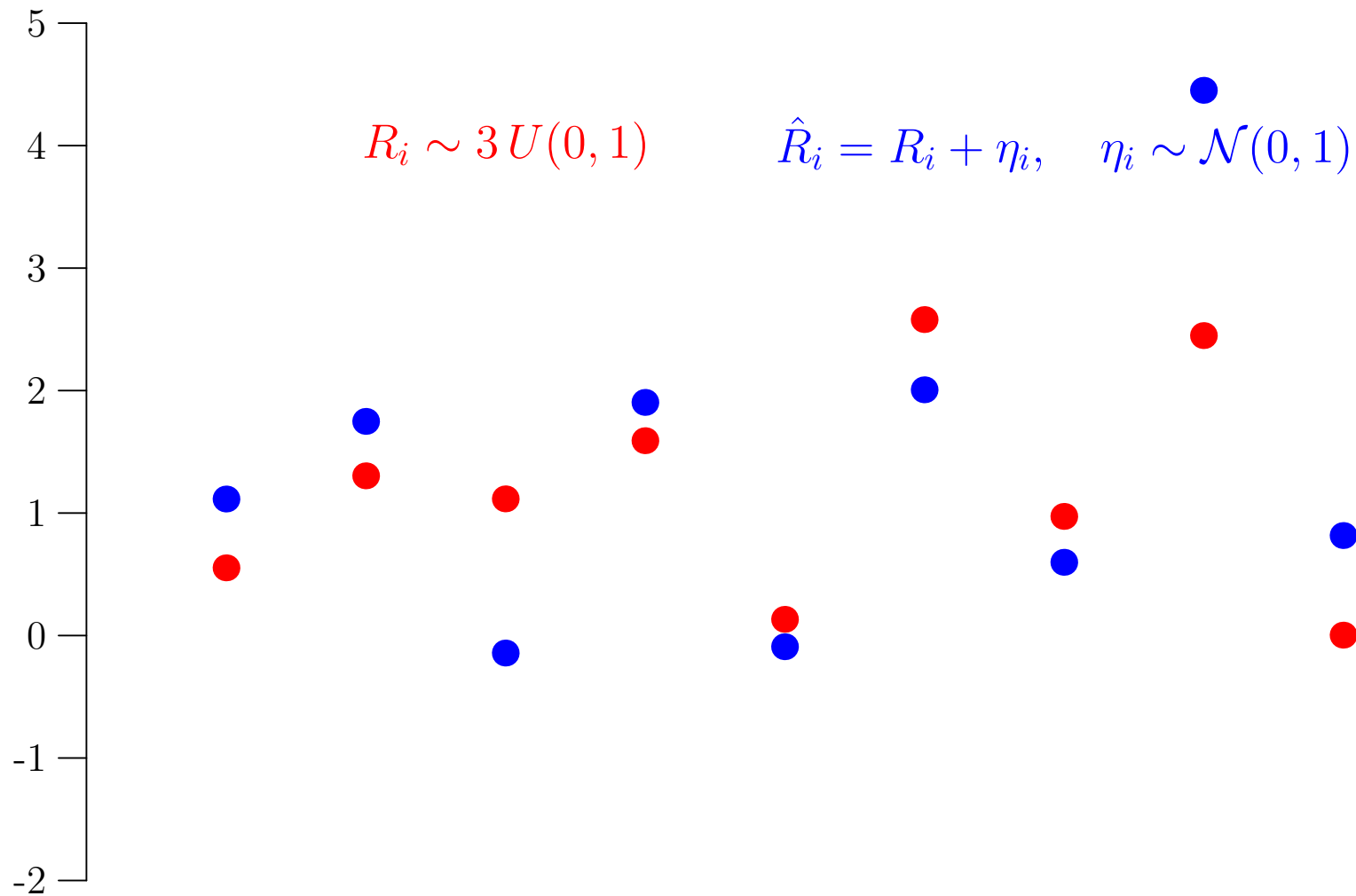
The Overfitting Game



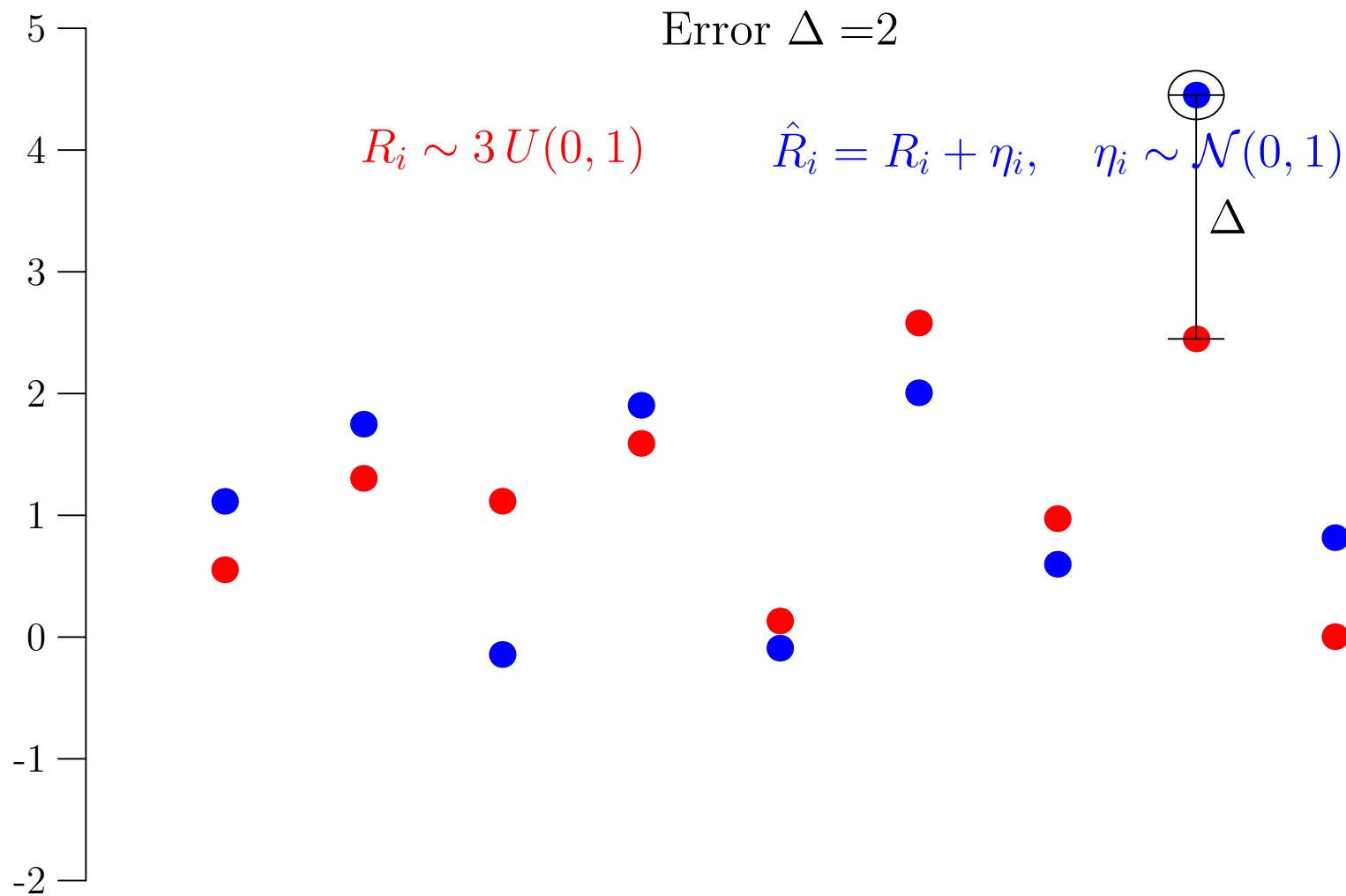
The Overfitting Game



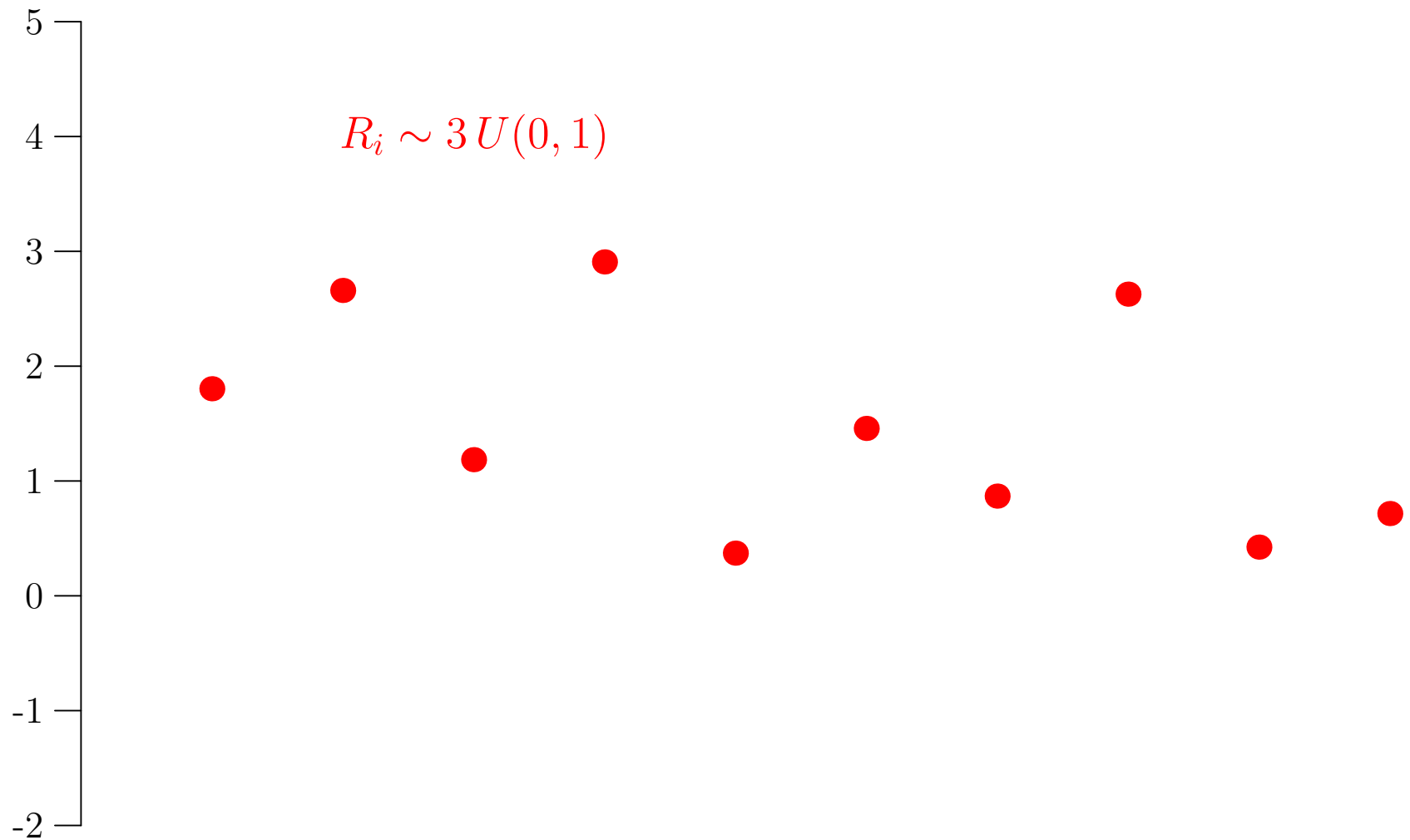
The Overfitting Game



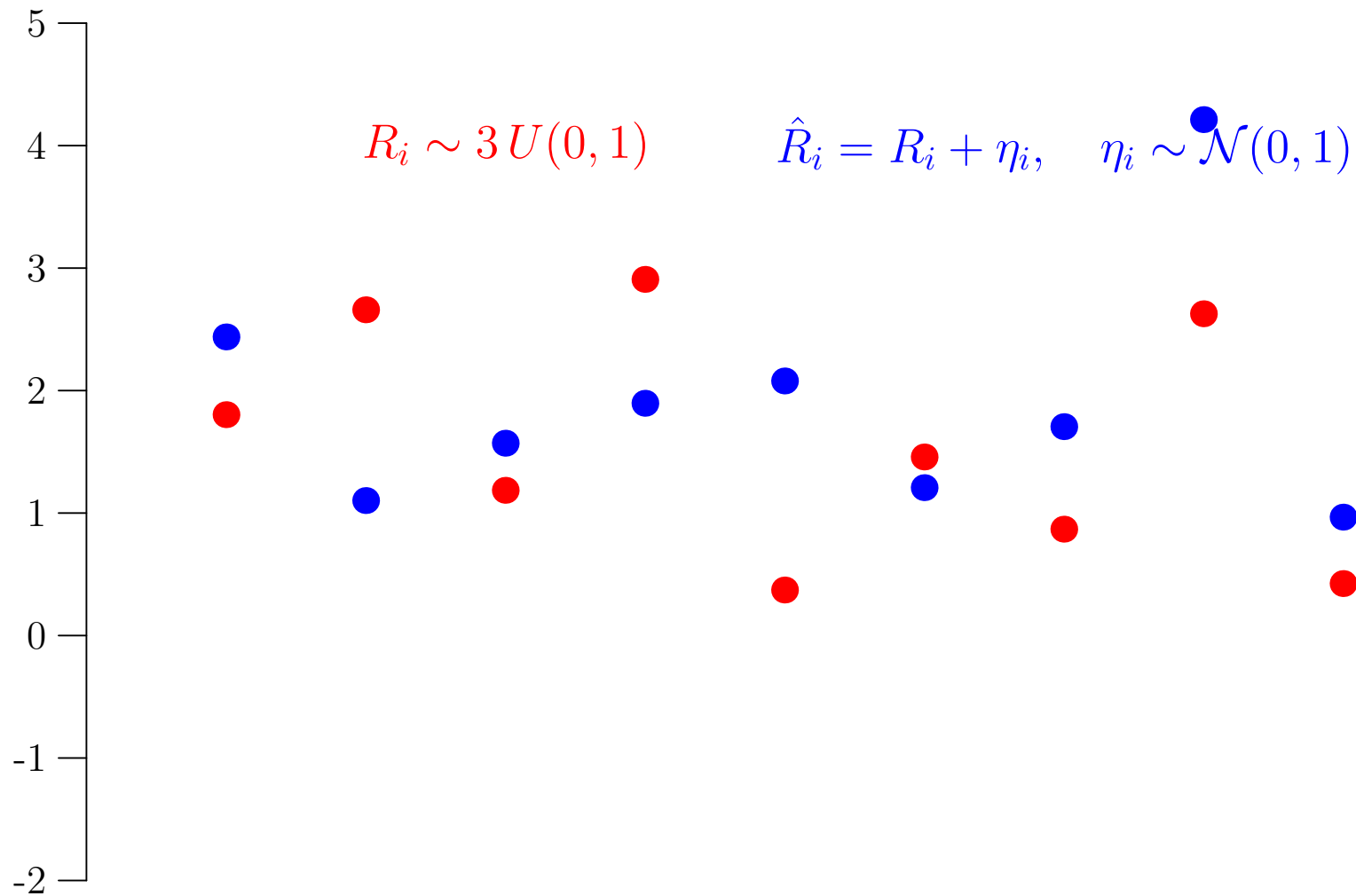
The Overfitting Game



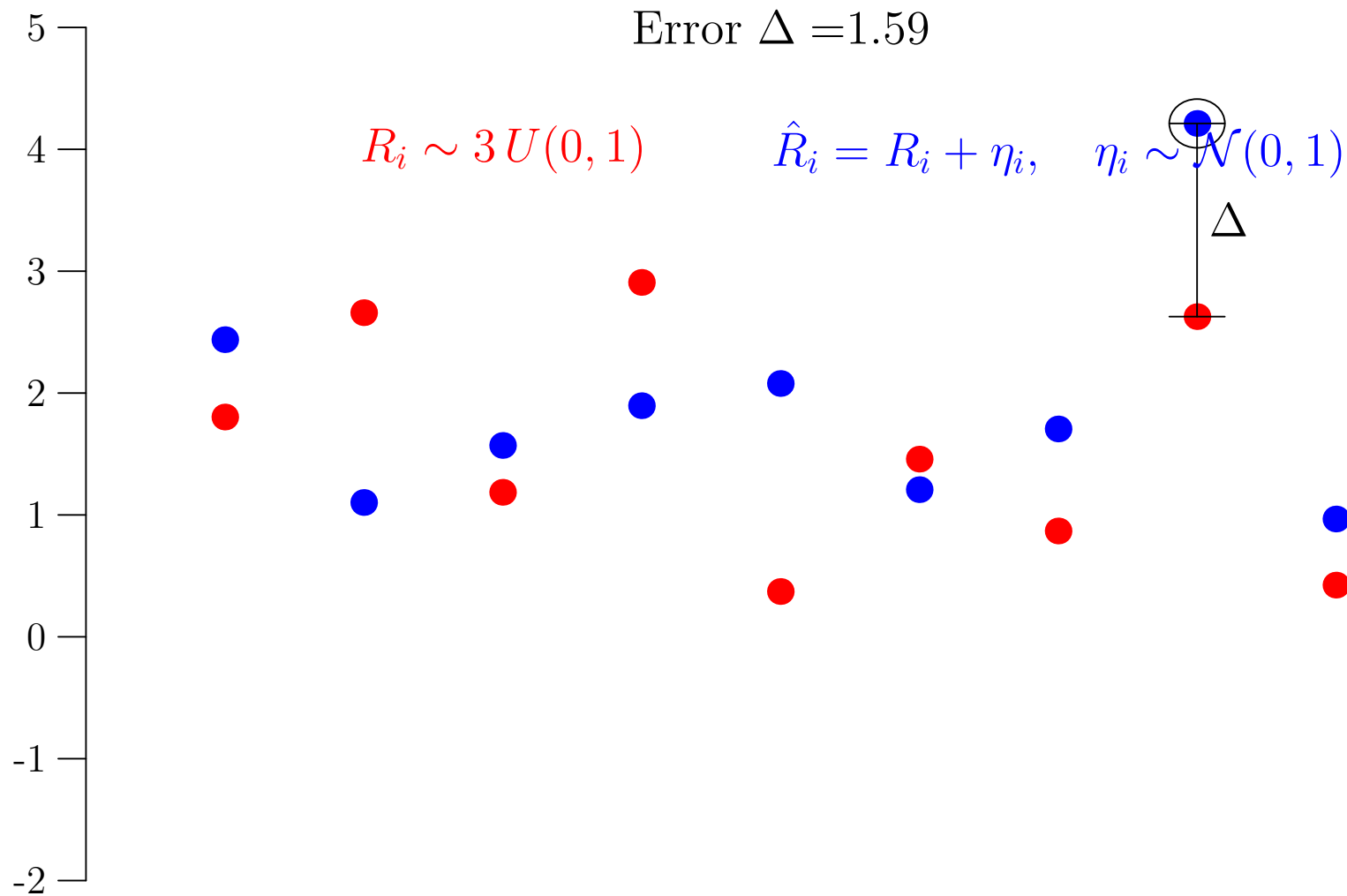
The Overfitting Game



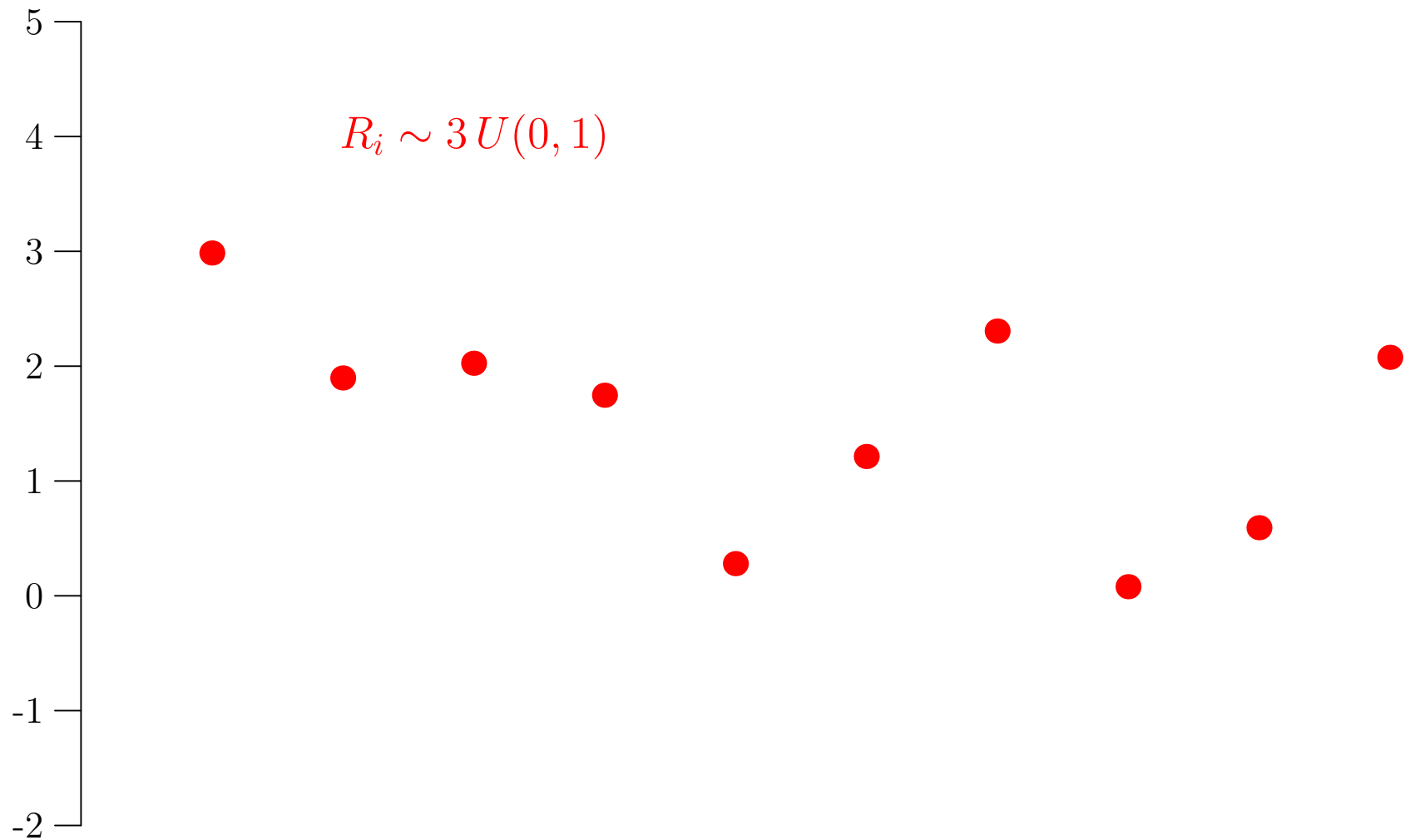
The Overfitting Game



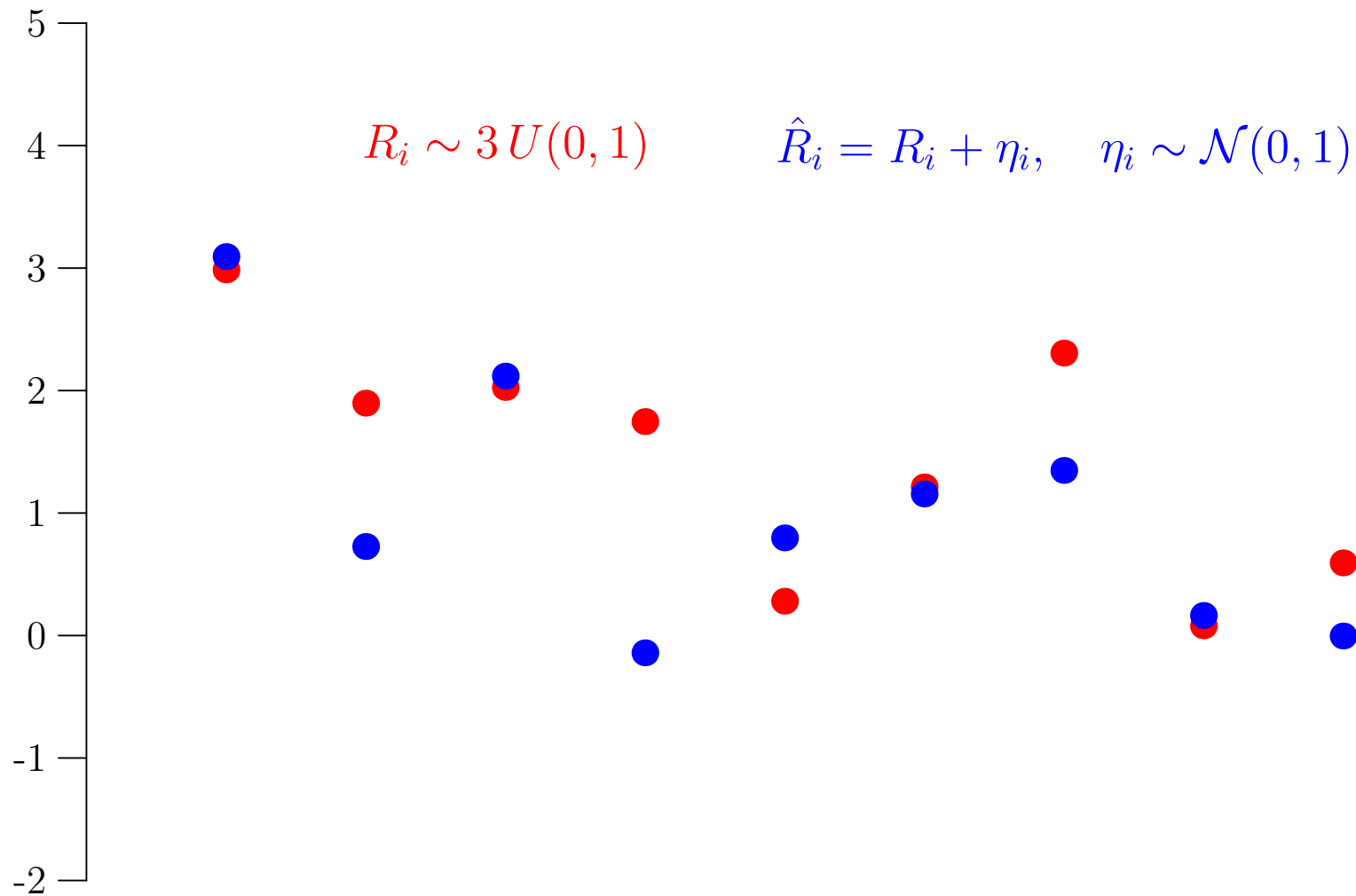
The Overfitting Game



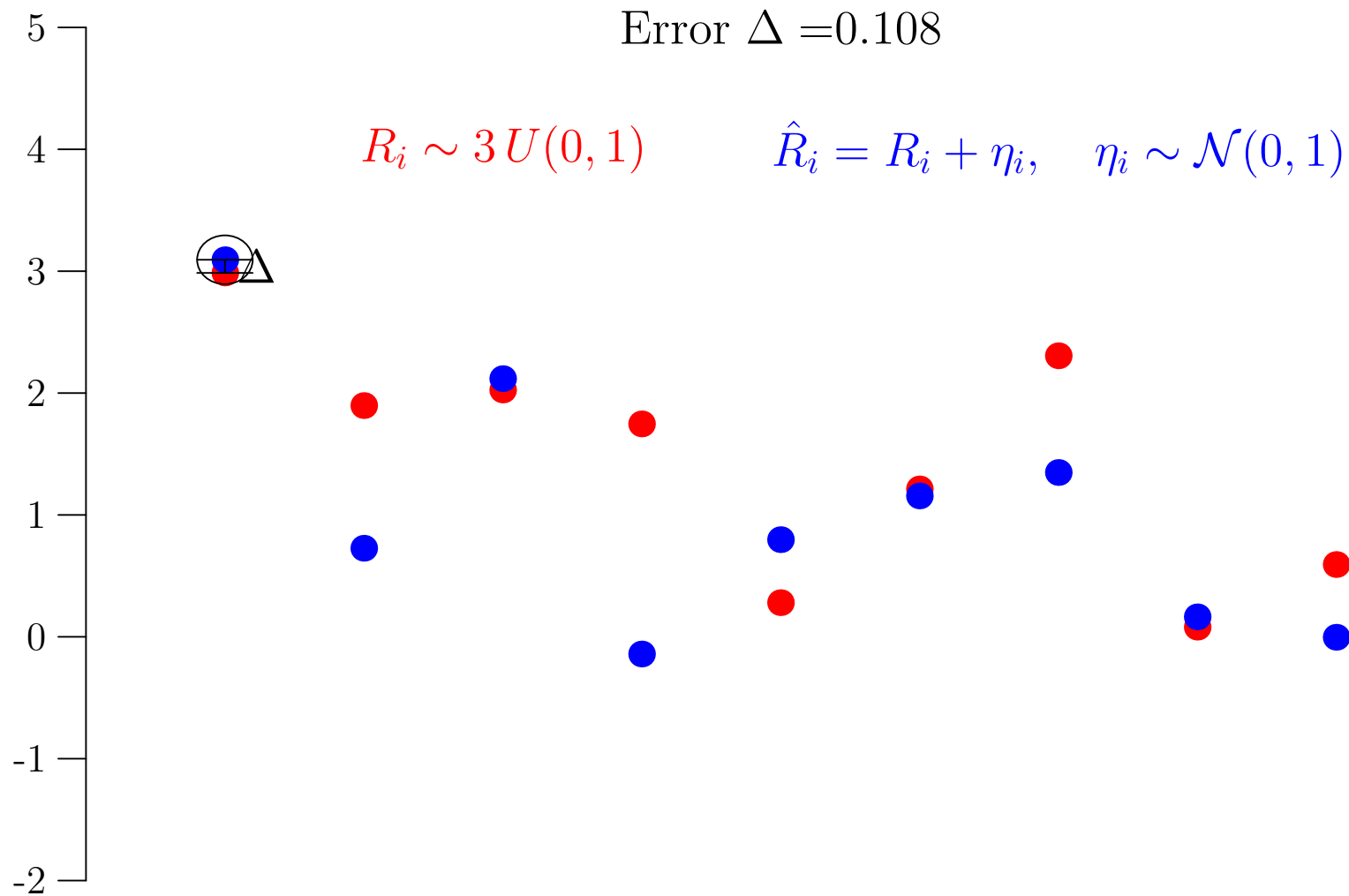
The Overfitting Game



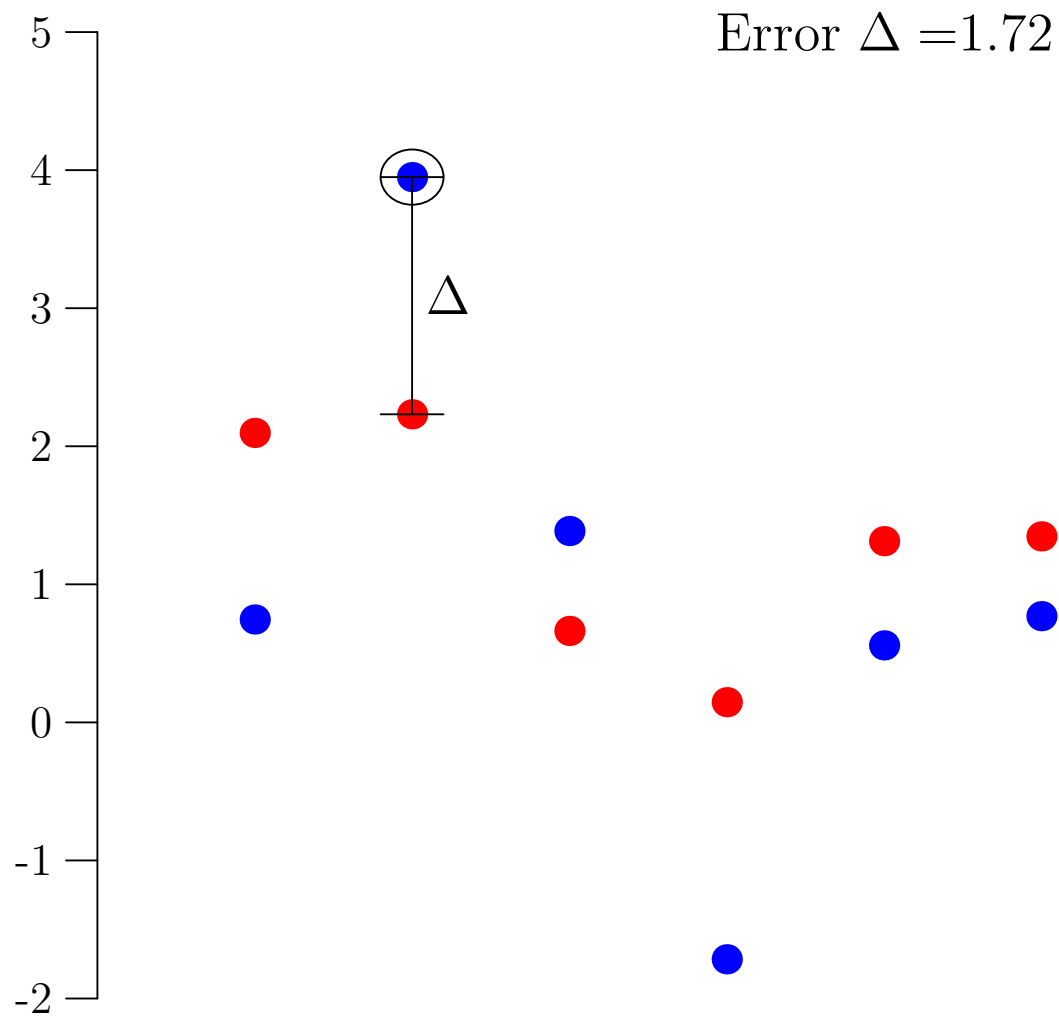
The Overfitting Game



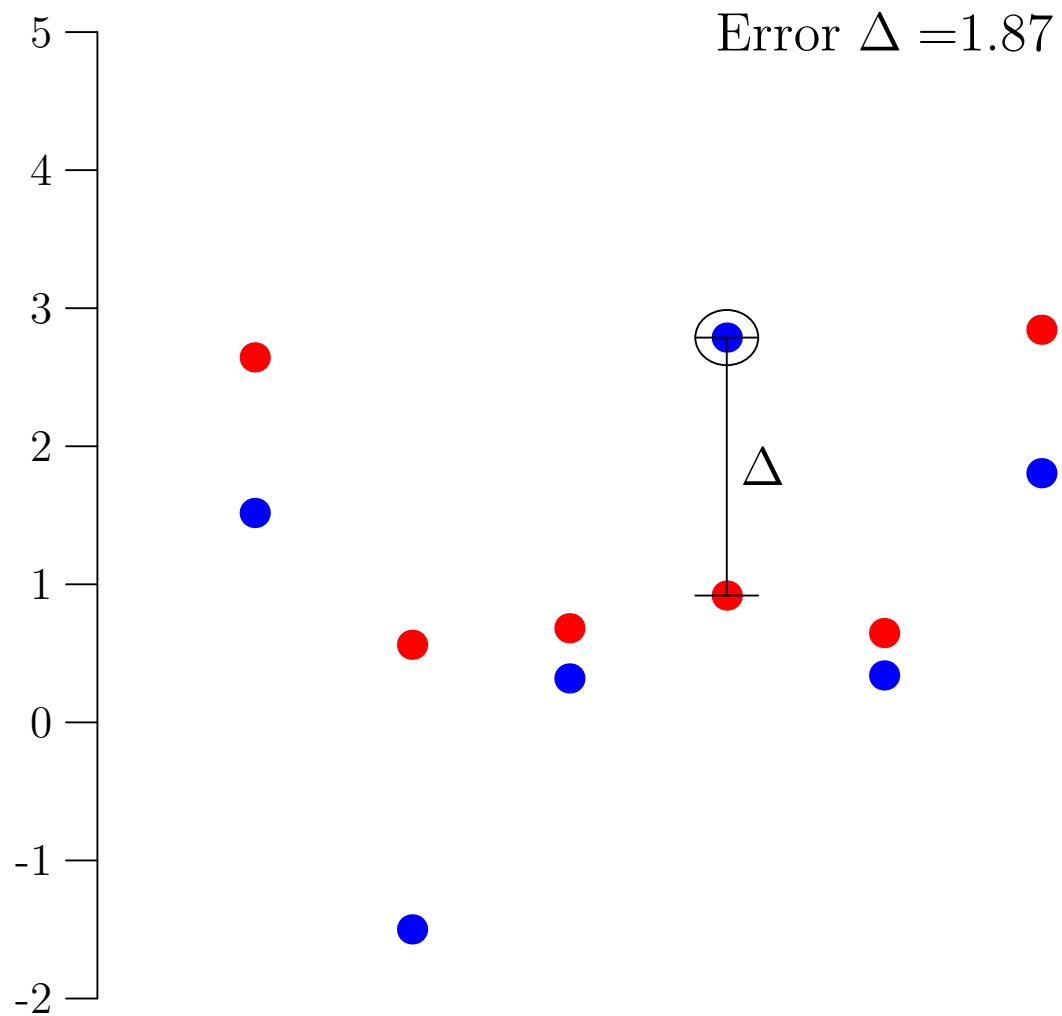
The Overfitting Game



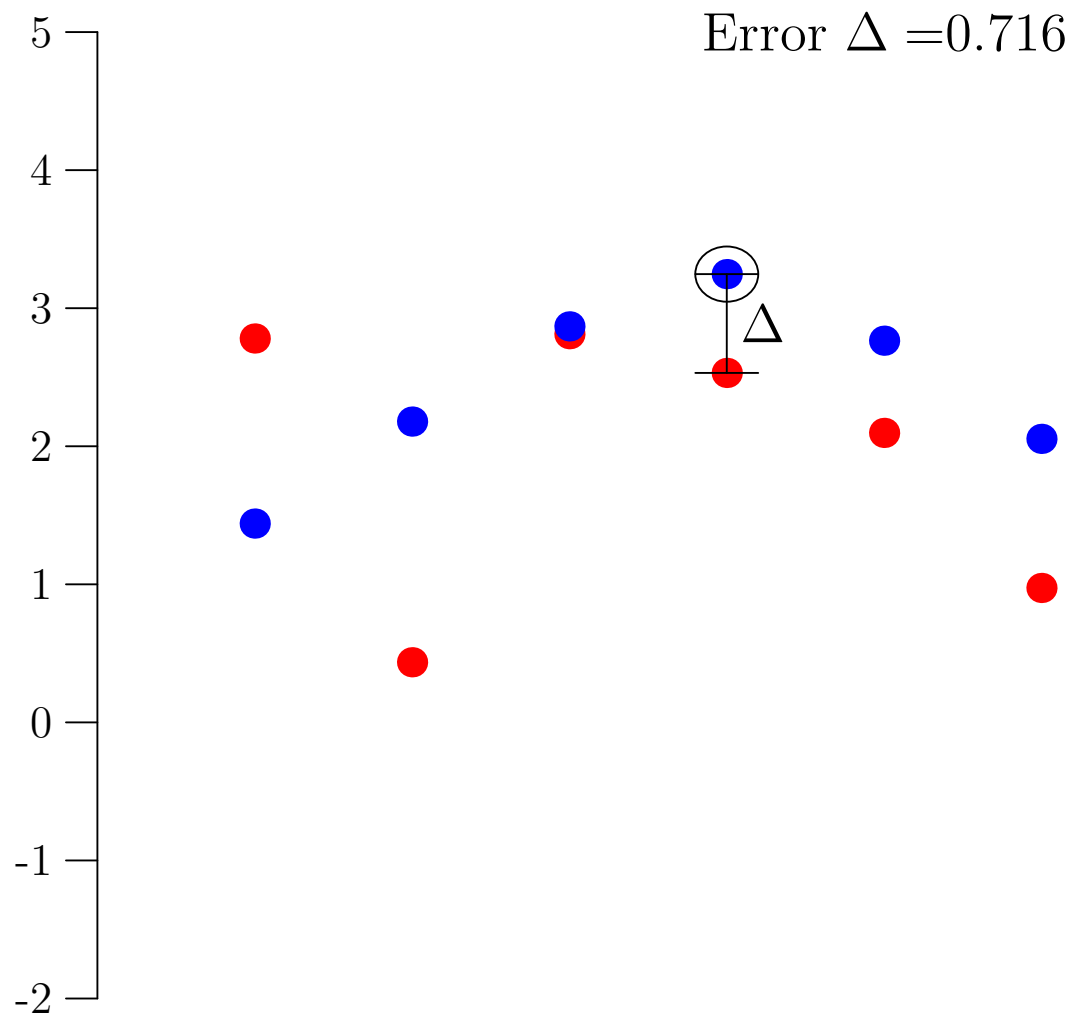
The Overfitting Game



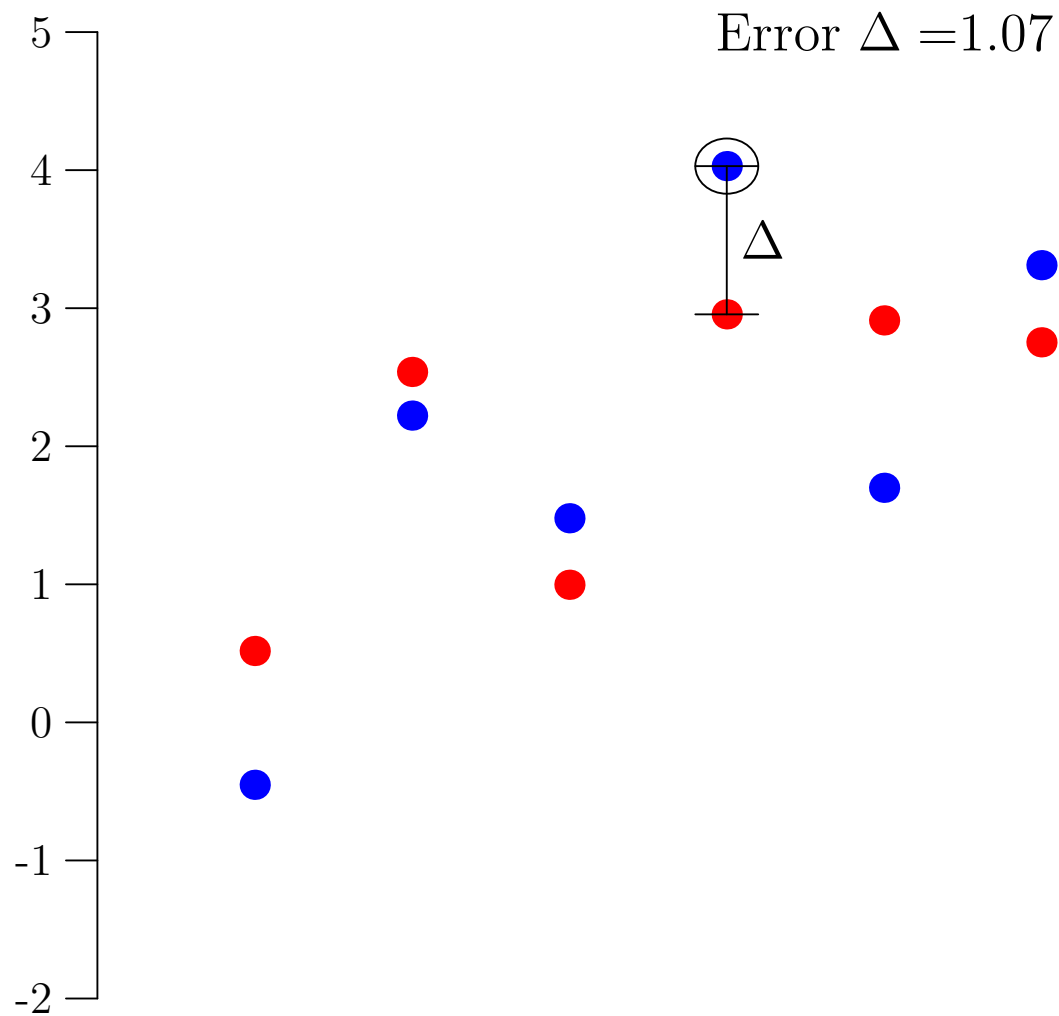
The Overfitting Game



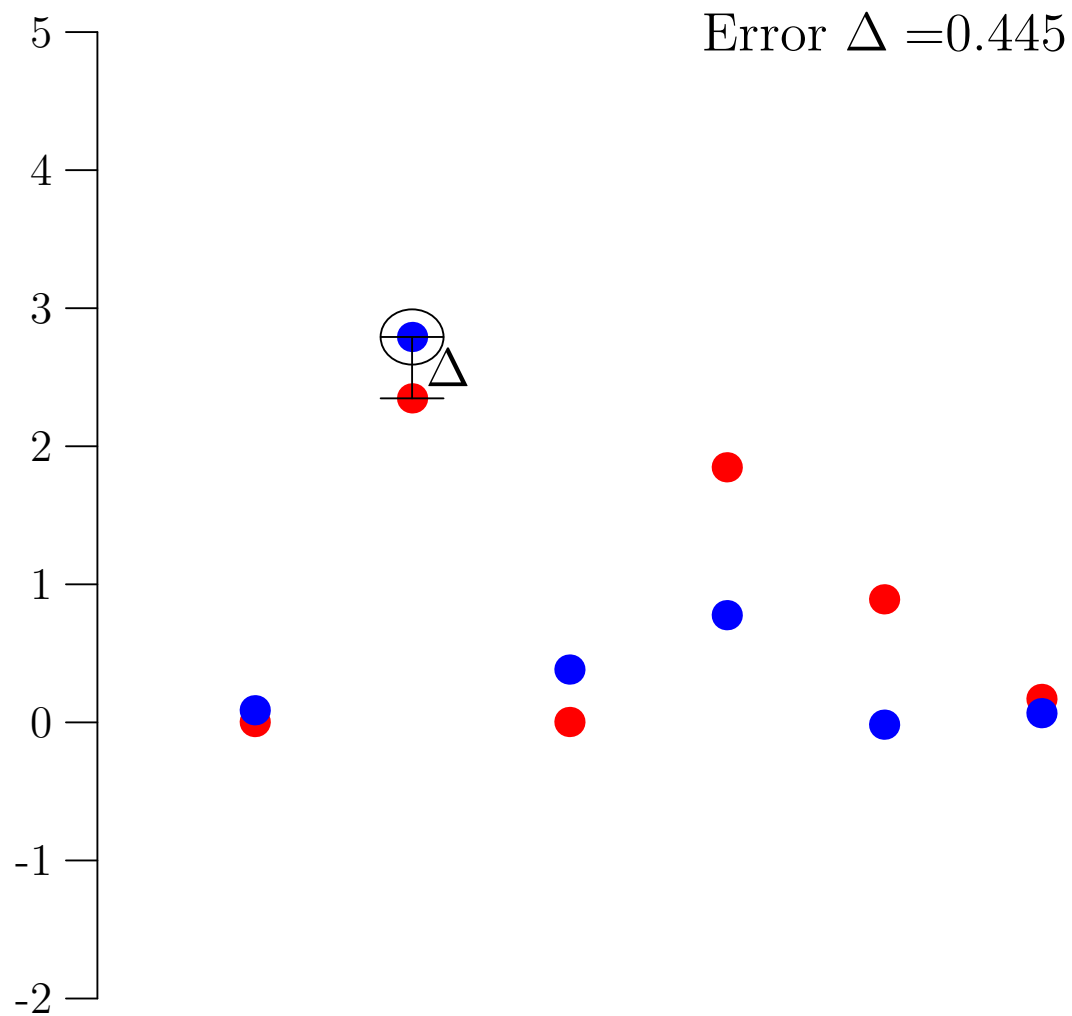
The Overfitting Game



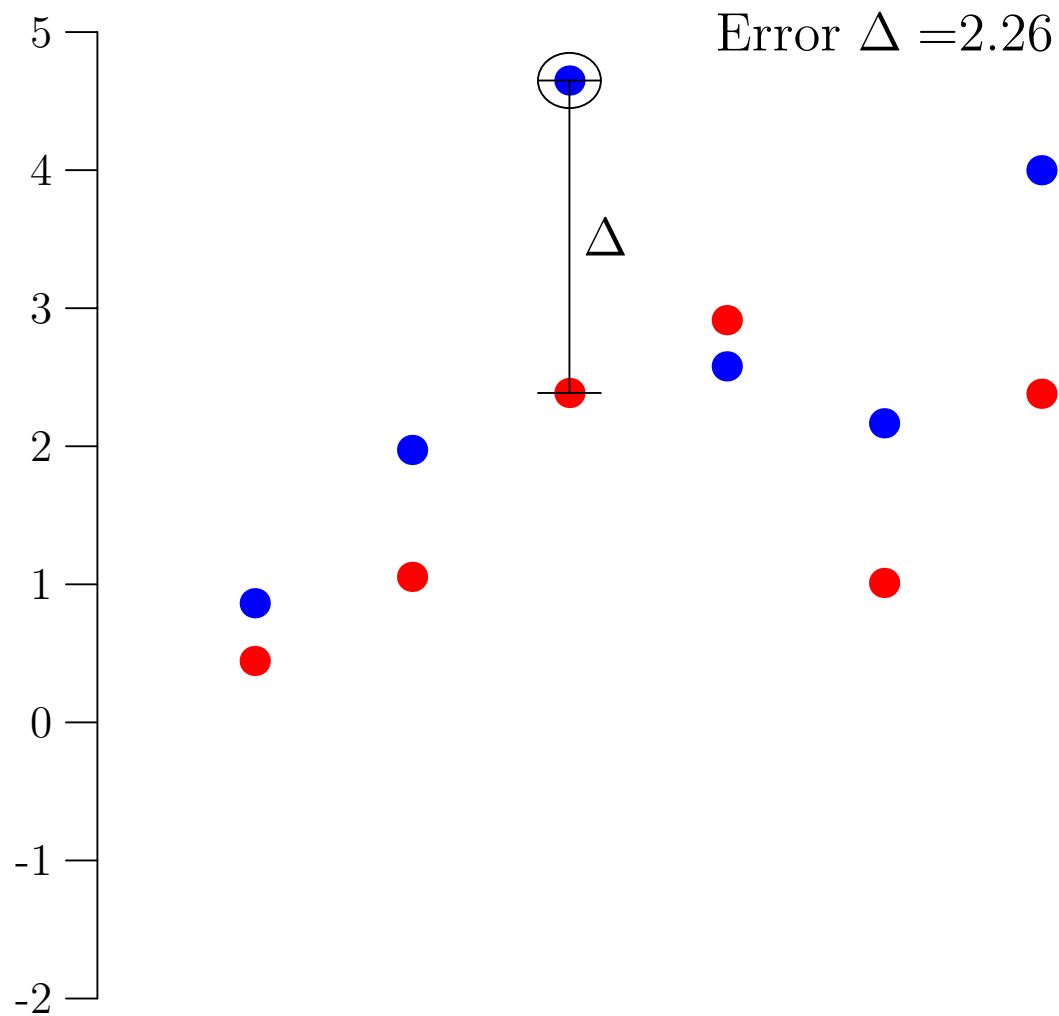
The Overfitting Game



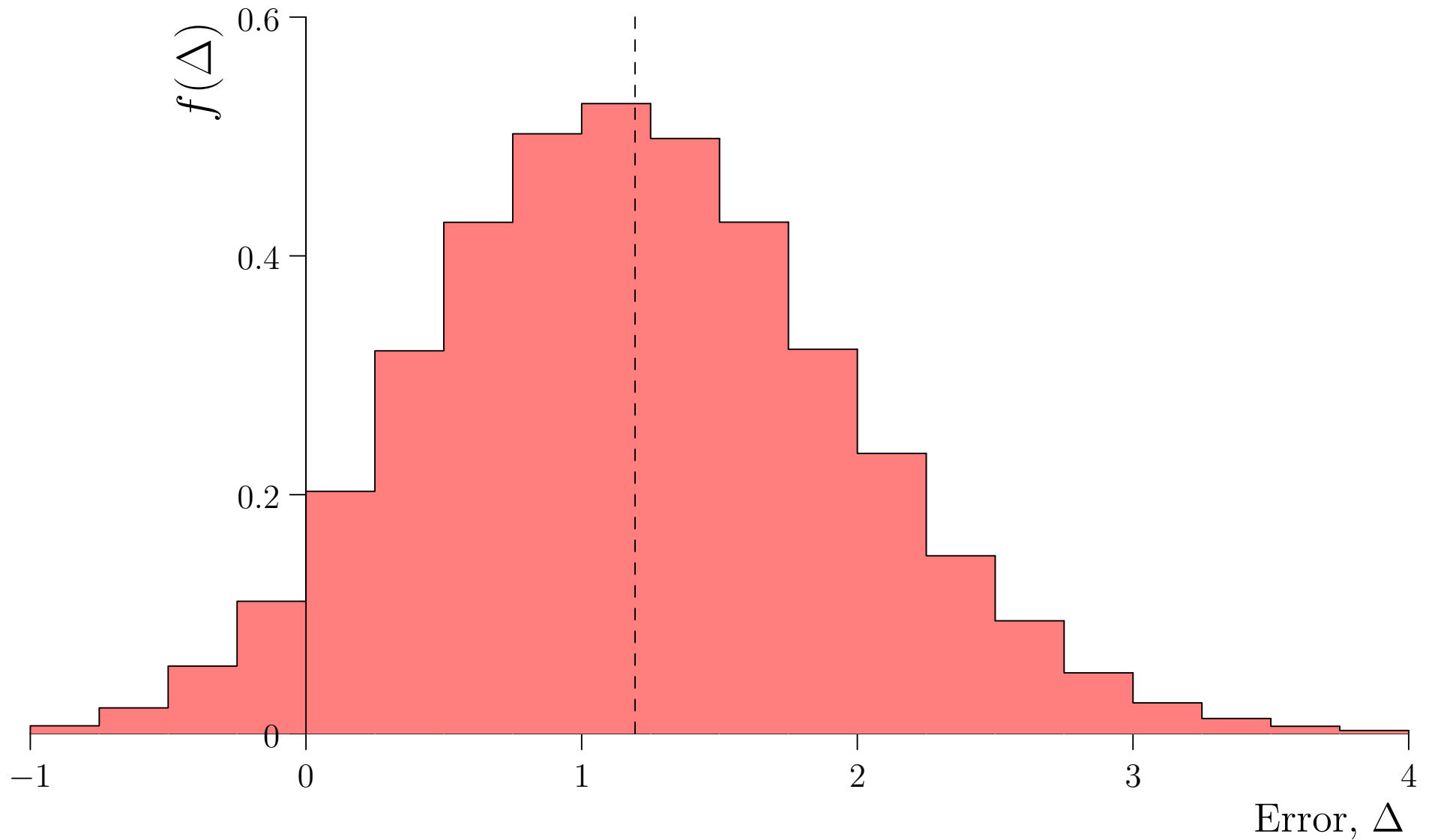
The Overfitting Game



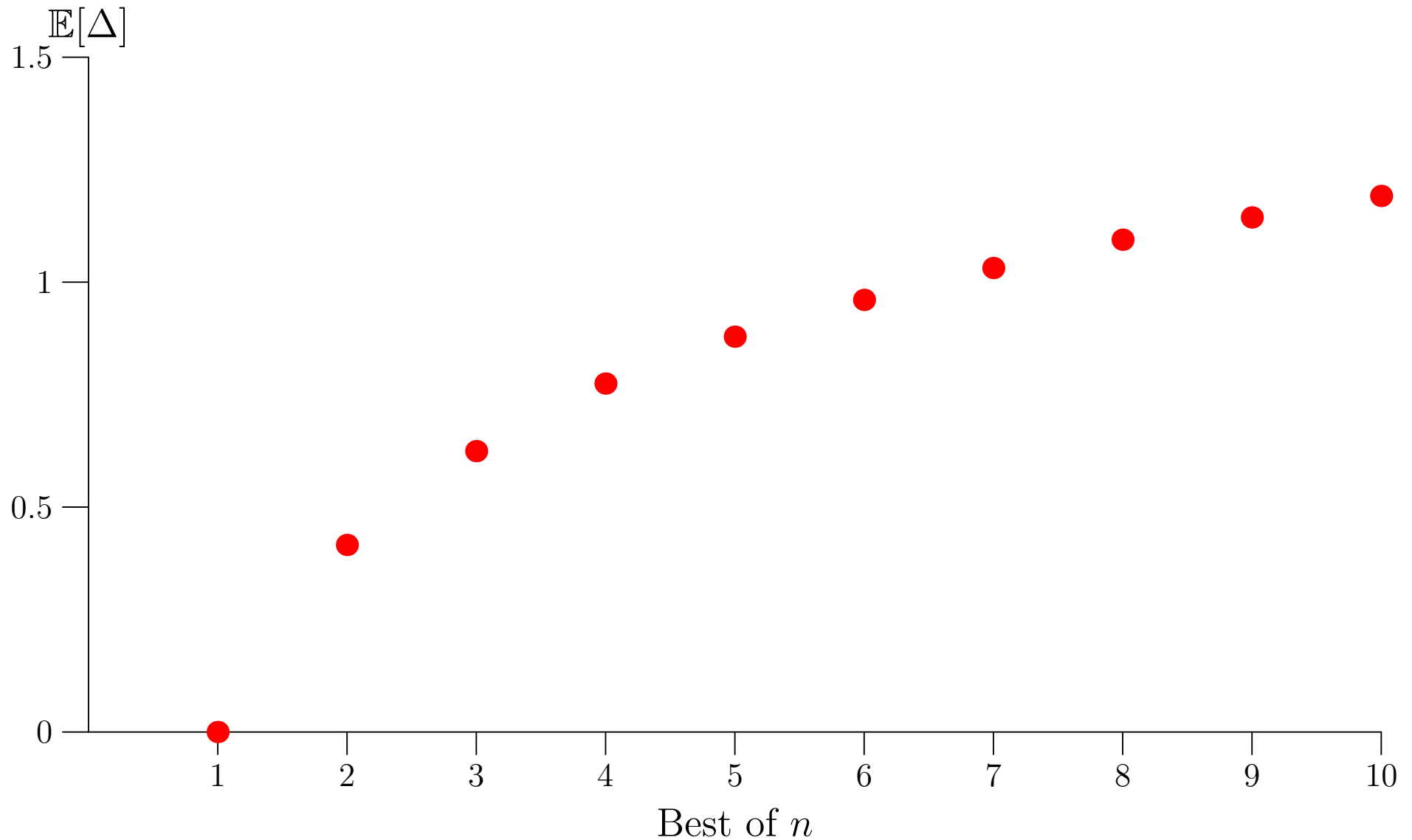
The Overfitting Game



The Overfitting Game



The Overfitting Game



Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test Set

Training Set

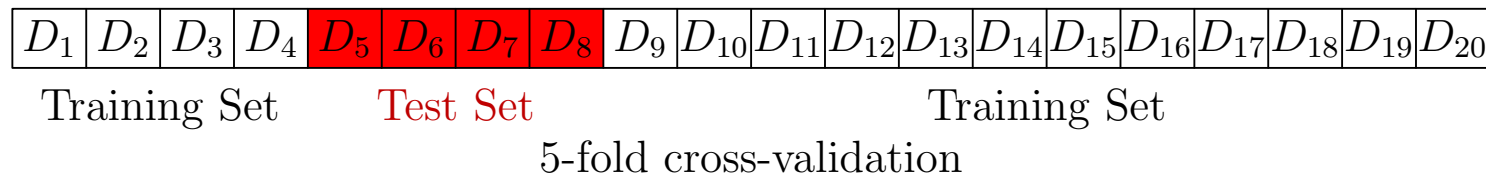
5-fold cross-validation

$$E_g = 5.1$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

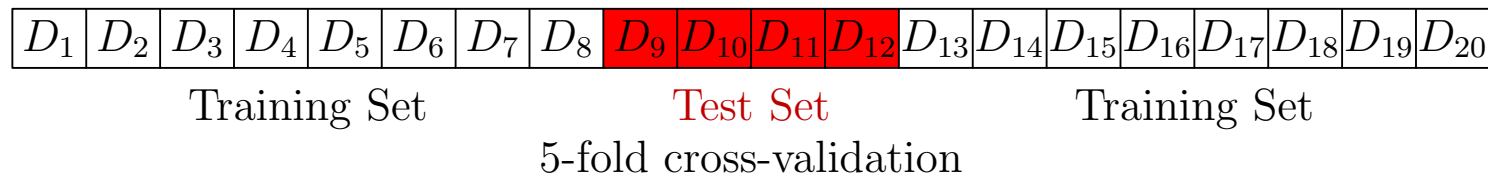


$$E_g = 3.7$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



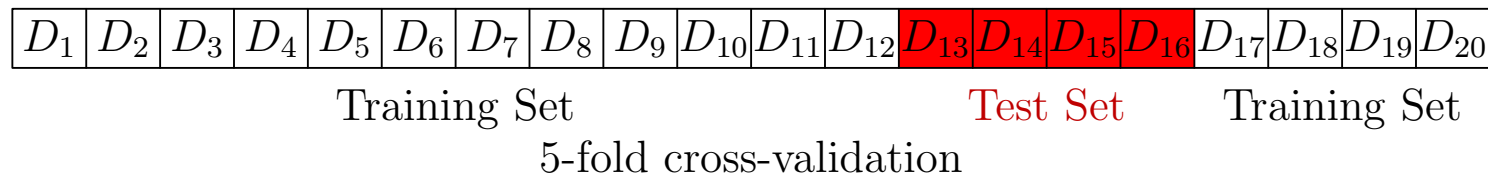
$E_g =$

4.6

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



$E_g =$

4.6

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Training Set

Test Set

5-fold cross-validation

$E_g =$

3.3

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

$$\langle E_g \rangle = \frac{5.1 + 3.7 + 4.6 + 4.6 + 3.3}{5} = 4.3$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test Set

Training Set

10-fold cross-validation

$$E_g = 5.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test Set

Training Set

10-fold cross-validation

$$E_g = 1.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Training Set Test Set

Training Set

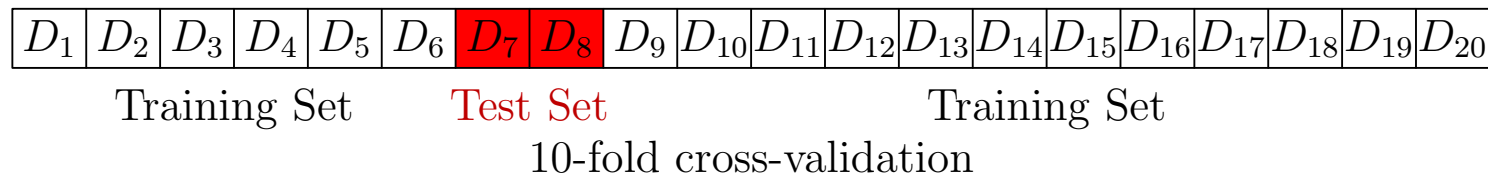
10-fold cross-validation

$$E_g = 4.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

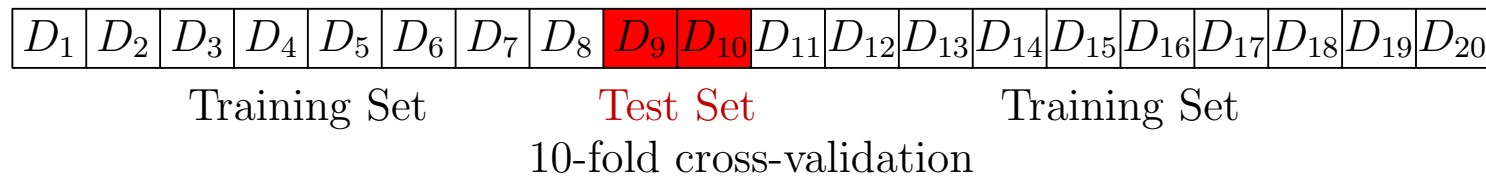


$$E_g = 3.6$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



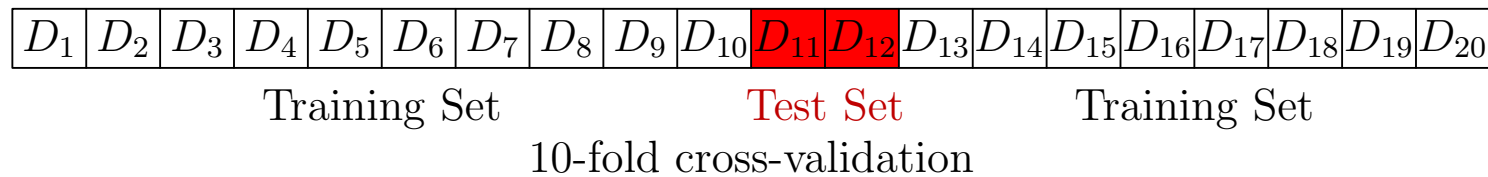
$E_g =$

7.4

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



$E_g =$

0.99

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Training Set

Test Set

Training Set

10-fold cross-validation

$E_g =$

4.5

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Training Set

Test Set

Training Set

10-fold cross-validation

$E_g =$

5.4

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Training Set

Test Set

10-fold cross-validation

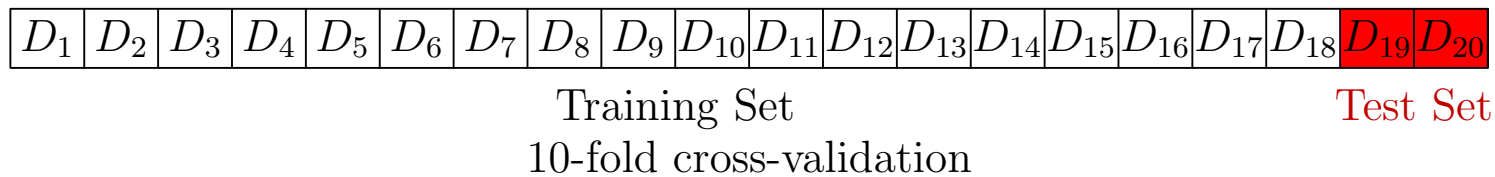
$E_g =$

6.2

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$



$E_g =$

2.7

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

$$\langle E_g \rangle = \frac{5.8 + 1.8 + 4.8 + 3.6 + 7.4 + 0.99 + 4.5 + 5.4 + 6.2 + 2.7}{10} = 4.3$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

$$E_g = 5$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

$$E_g = 3.7$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

$$E_g = 5$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

$$E_g = 2.2$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

Test

Leave-one-out cross-validation

$$E_g = 3.8$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

$$\langle E_g \rangle = 3.9$$

Cross Validation

- If you want to use more data for training then you can use cross validation
- K -fold cross validation splits the data into K groups

$$\mathcal{D} = \{D_i\}_{i=1}^P \quad D_i = (\mathbf{x}_i, y_i)$$

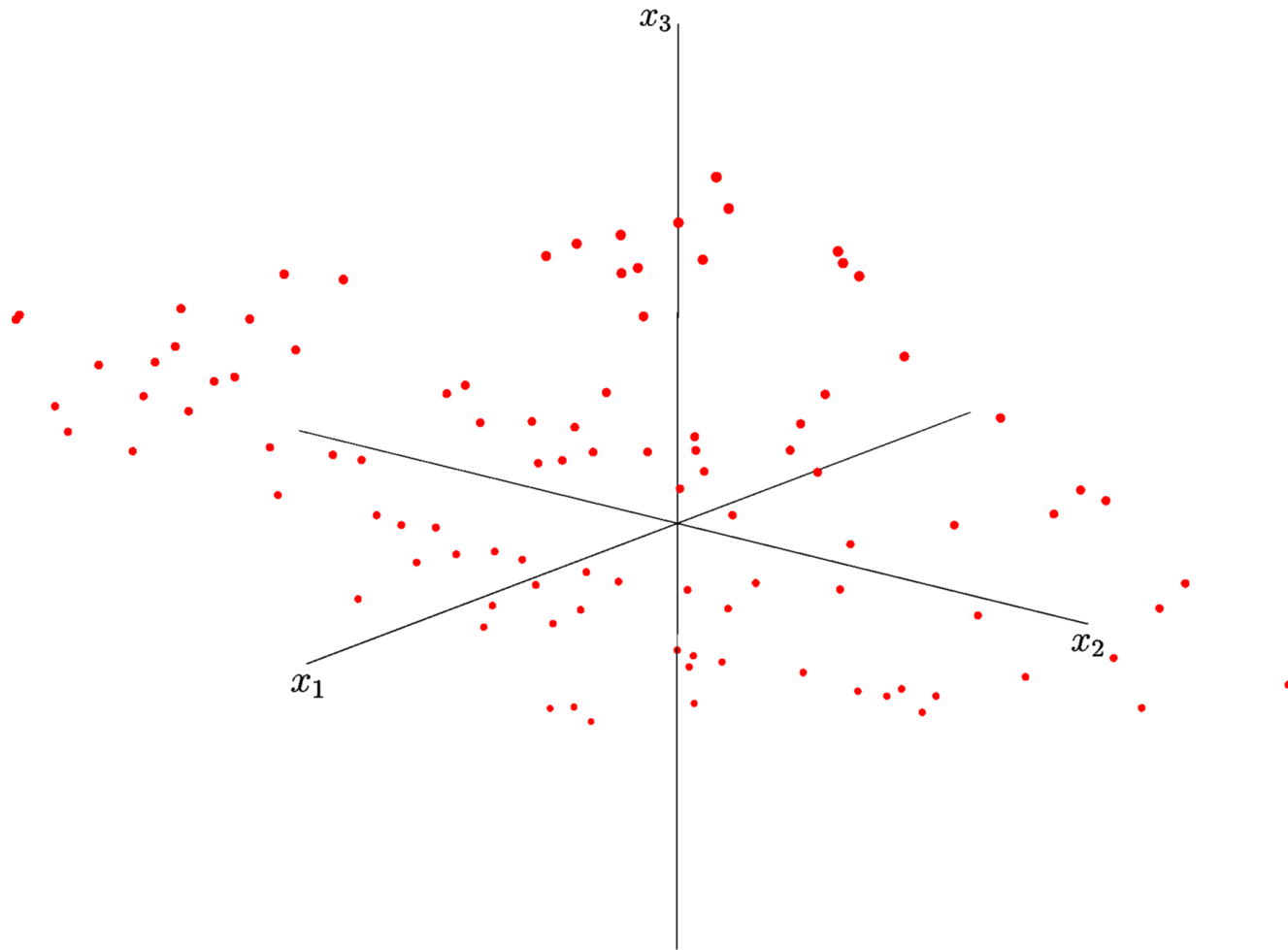
D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}	D_{16}	D_{17}	D_{18}	D_{19}	D_{20}
-------	-------	-------	-------	-------	-------	-------	-------	-------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------	----------

$$\langle E_g \rangle = 3.9$$

- Leave-one-out cross-validation is extreme case

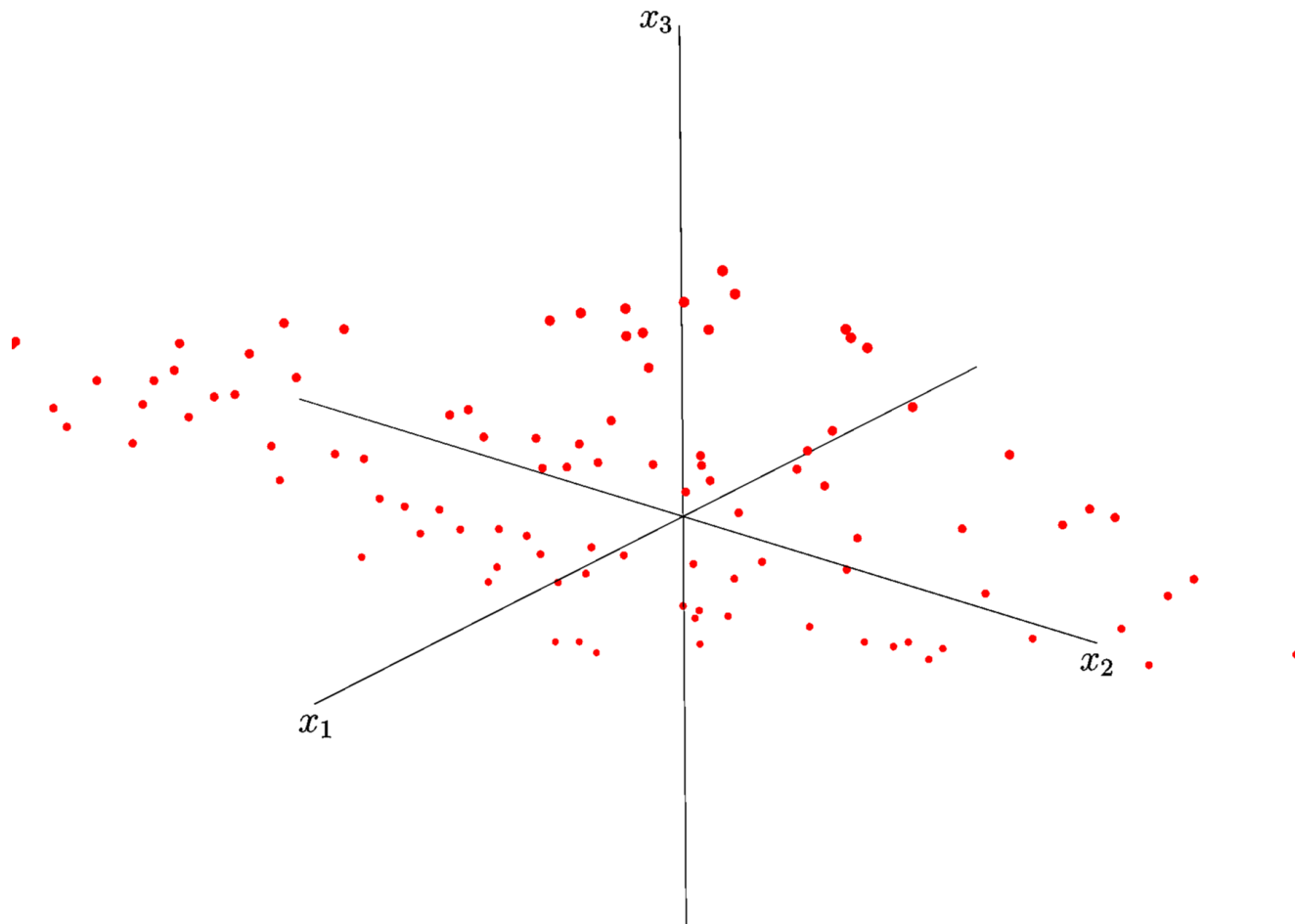
Hidden Structure

Can't spot low dimensional data by looking at numbers



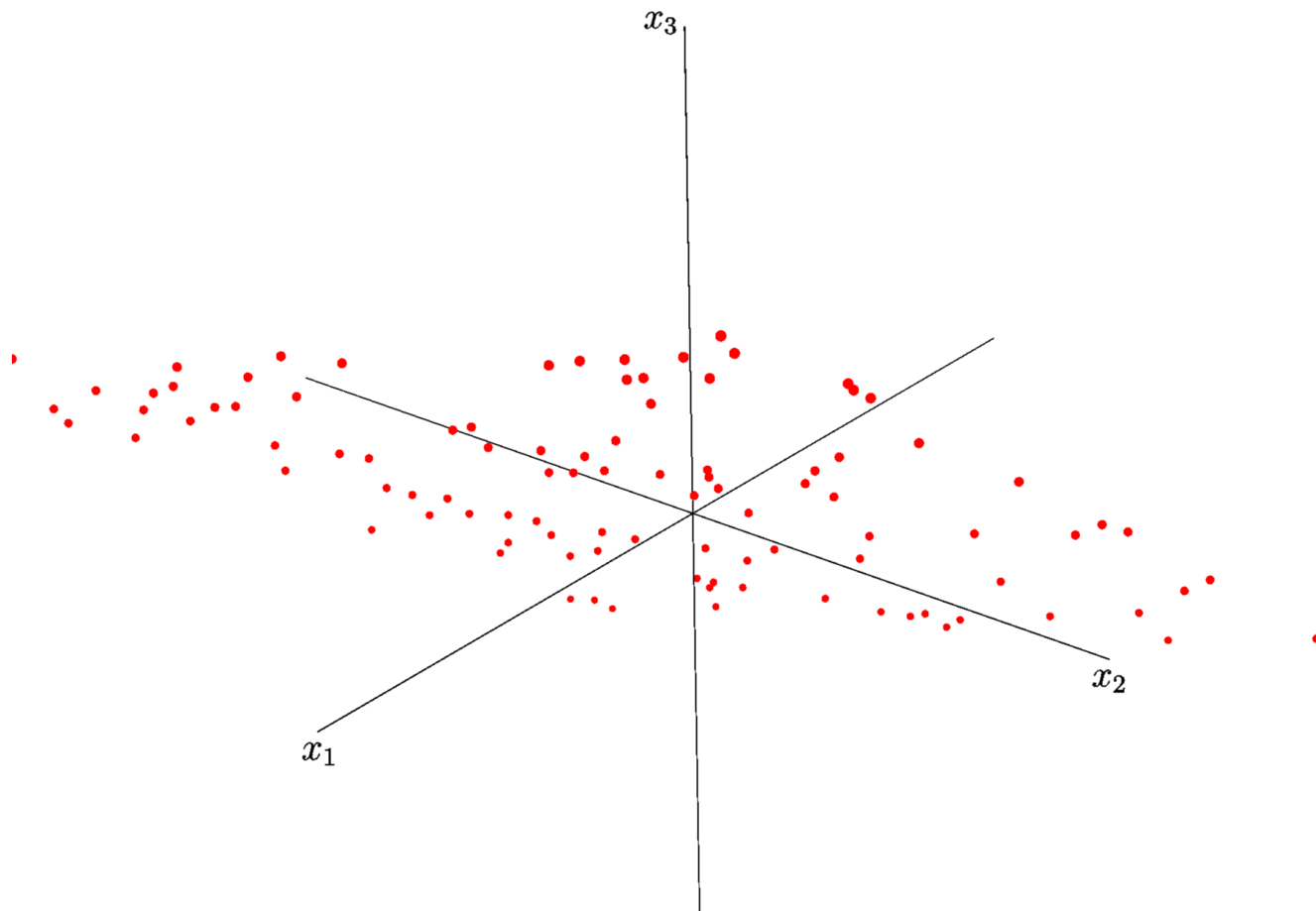
Hidden Structure

Can't spot low dimensional data by looking at numbers



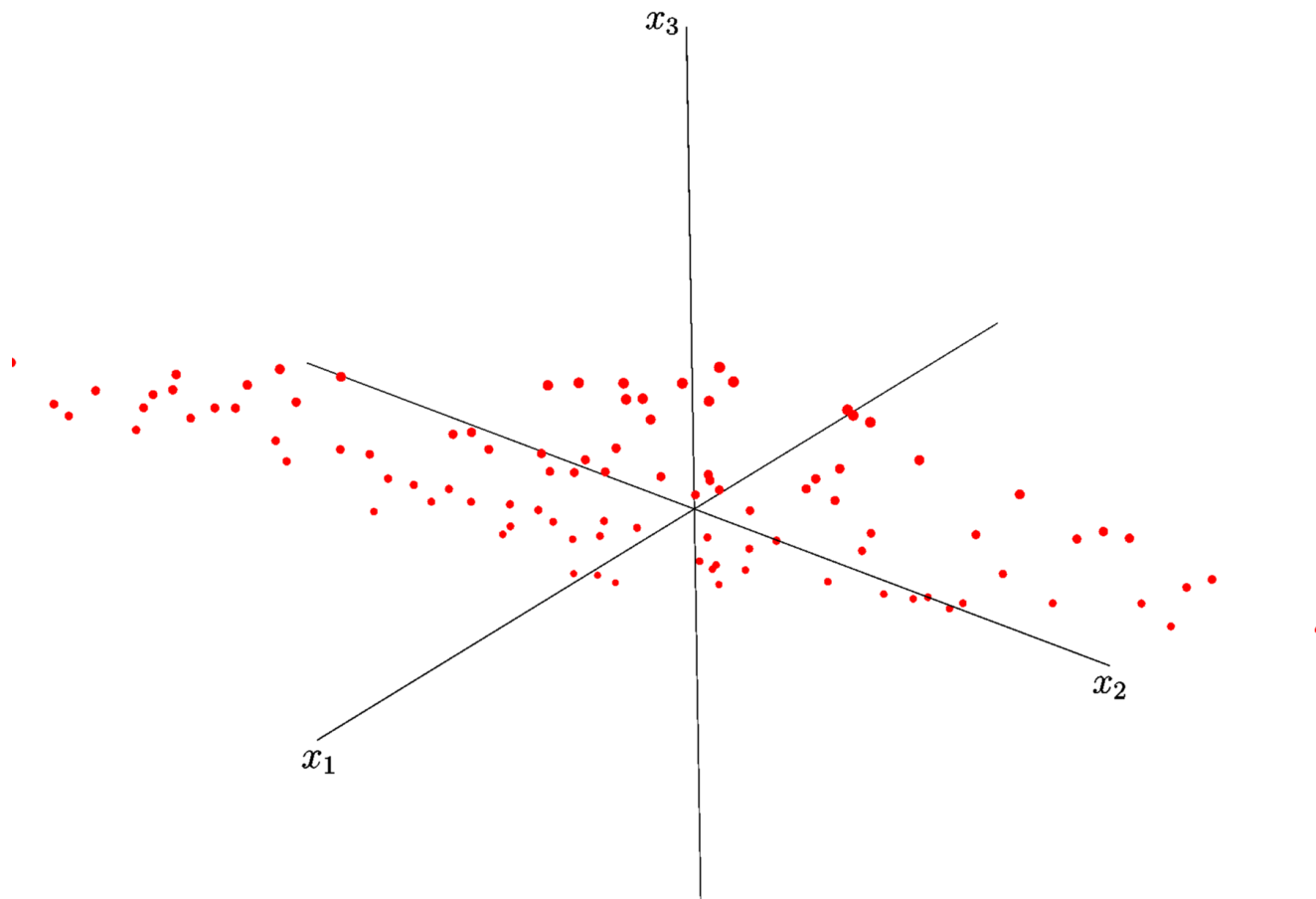
Hidden Structure

Can't spot low dimensional data by looking at numbers



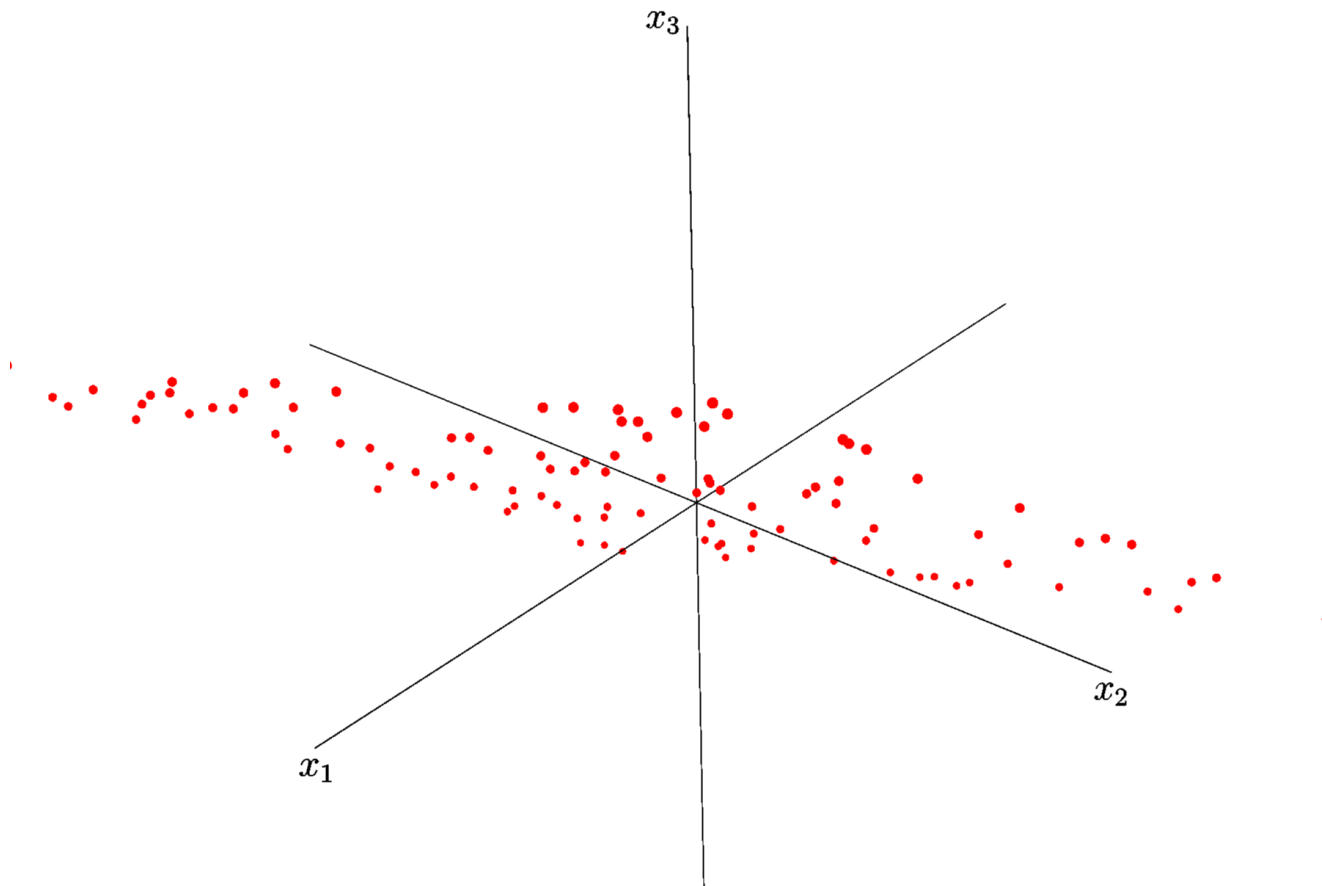
Hidden Structure

Can't spot low dimensional data by looking at numbers



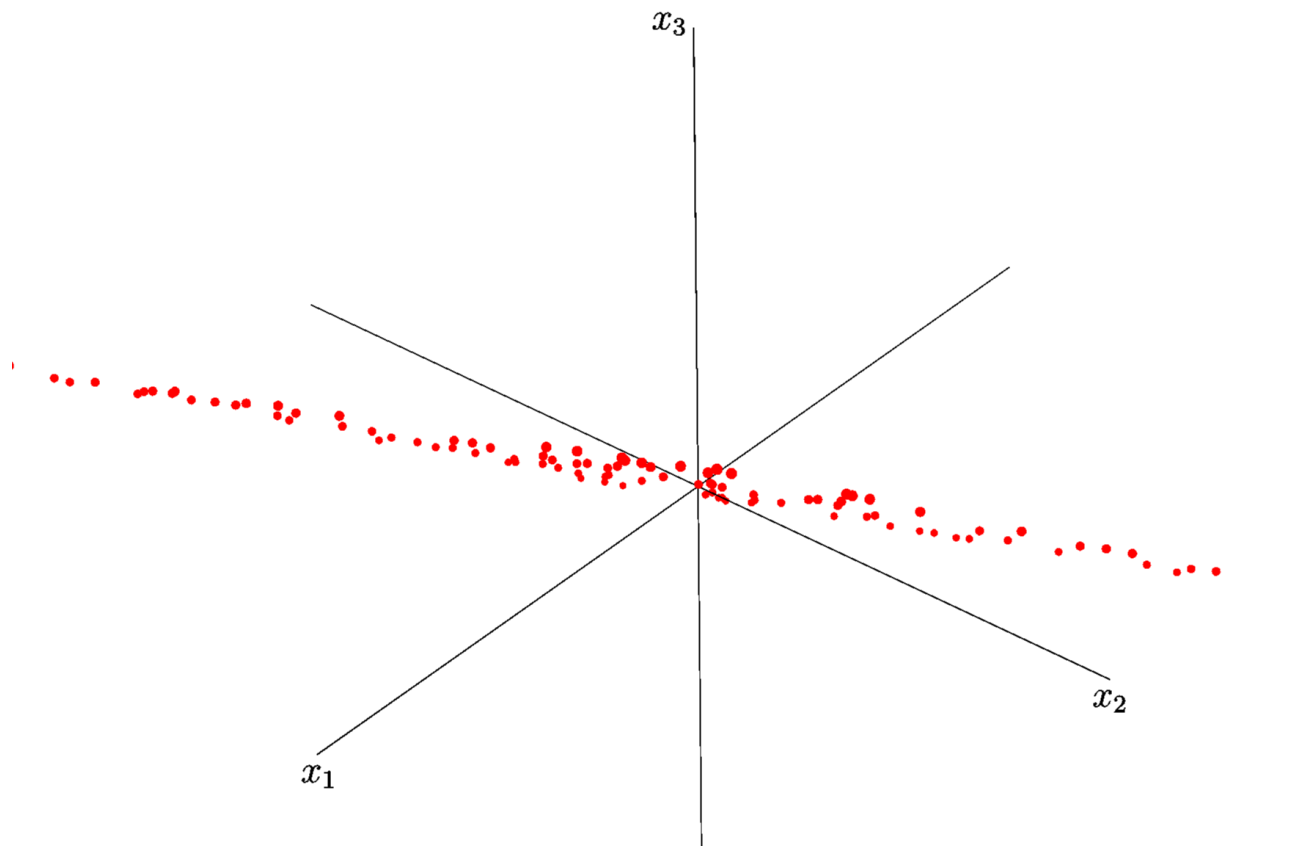
Hidden Structure

Can't spot low dimensional data by looking at numbers



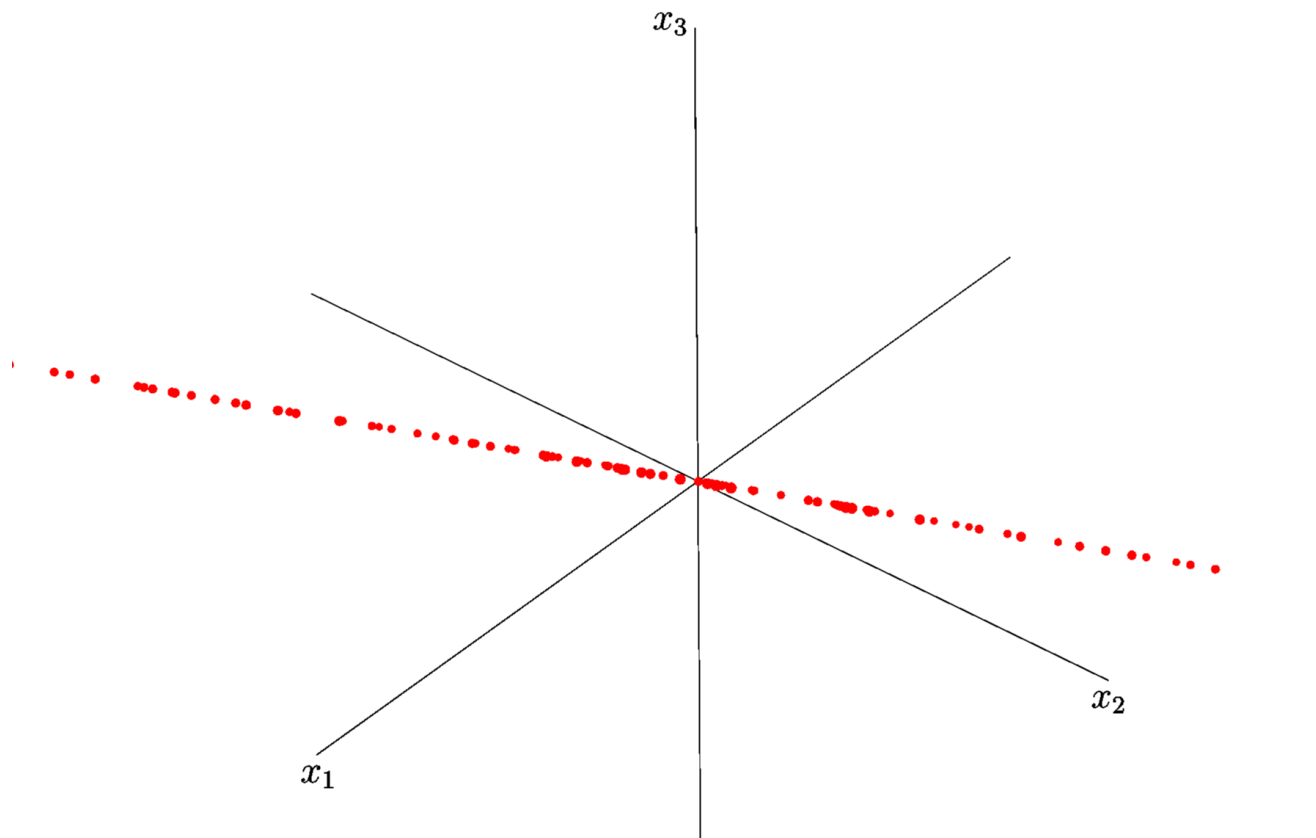
Hidden Structure

Can't spot low dimensional data by looking at numbers



Hidden Structure

Can't spot low dimensional data by looking at numbers



Dimensionality Reduction

- We can sometimes simplify our machines by using less features
- We can project our data onto a lower dimensional sub-space (e.g. one with the maximum variation in the data: PCA)
- We can use clustering to find exemplars and recode our data in terms of distances from the exemplars (radial basis functions)
- Whether this helps depends on whether the information we discard is pertinent to the task we are trying to perform

Dimensionality Reduction

- We can sometimes simplify our machines by using less features
- We can project our data onto a lower dimensional sub-space (e.g. one with the maximum variation in the data: PCA)
- We can use clustering to find exemplars and recode our data in terms of distances from the exemplars (radial basis functions)
- Whether this helps depends on whether the information we discard is pertinent to the task we are trying to perform

Dimensionality Reduction

- We can sometimes simplify our machines by using less features
- We can project our data onto a lower dimensional sub-space (e.g. one with the maximum variation in the data: PCA)
- We can use clustering to find exemplars and recode our data in terms of distances from the exemplars (radial basis functions)
- Whether this helps depends on whether the information we discard is pertinent to the task we are trying to perform

Dimensionality Reduction

- We can sometimes simplify our machines by using less features
- We can project our data onto a lower dimensional sub-space (e.g. one with the maximum variation in the data: PCA)
- We can use clustering to find exemplars and recode our data in terms of distances from the exemplars (radial basis functions)
- Whether this helps depends on whether the information we discard is pertinent to the task we are trying to perform

Feature Selection

- Spurious features will allow us to find spurious rules (**over-fitting**)
- We can try different combinations of features to find the best set, although it rapidly becomes intractable to do this in all ways
- We can use various heuristics to decide which features to keep, but no heuristic is fail-safe method to find the best set of features
- Feature selection however can be powerful, often we can get very good results by keeping only a few variables
- As well as possibly improving generalisation we also get a more **interpretable** rule

Feature Selection

- Spurious features will allow us to find spurious rules (**over-fitting**)
- We can try different combinations of features to find the best set, although it rapidly becomes intractable to do this in all ways
- We can use various heuristics to decide which features to keep, but no heuristic is fail-safe method to find the best set of features
- Feature selection however can be powerful, often we can get very good results by keeping only a few variables
- As well as possibly improving generalisation we also get a more **interpretable** rule

Feature Selection

- Spurious features will allow us to find spurious rules (**over-fitting**)
- We can try different combinations of features to find the best set, although it rapidly becomes intractable to do this in all ways
- We can use various heuristics to decide which features to keep, but no heuristic is fail-safe method to find the best set of features
- Feature selection however can be powerful, often we can get very good results by keeping only a few variables
- As well as possibly improving generalisation we also get a more **interpretable** rule

Feature Selection

- Spurious features will allow us to find spurious rules (**over-fitting**)
- We can try different combinations of features to find the best set, although it rapidly becomes intractable to do this in all ways
- We can use various heuristics to decide which features to keep, but no heuristic is fail-safe method to find the best set of features
- Feature selection however can be powerful, often we can get very good results by keeping only a few variables
- As well as possibly improving generalisation we also get a more **interpretable** rule

Feature Selection

- Spurious features will allow us to find spurious rules (**over-fitting**)
- We can try different combinations of features to find the best set, although it rapidly becomes intractable to do this in all ways
- We can use various heuristics to decide which features to keep, but no heuristic is fail-safe method to find the best set of features
- Feature selection however can be powerful, often we can get very good results by keeping only a few variables
- As well as possibly improving generalisation we also get a more **interpretable** rule

Normalising Features

- Measuring a feature in millimeters or kilometers is going to make a lot of difference to the size of that feature
- Many learning algorithms are sensitive to the size of a feature (larger features are more important)
- If we don't know how important different features are then it makes sense to normalise our features

Normalising Features

- Measuring a feature in millimeters or kilometers is going to make a lot of difference to the size of that feature
- Many learning algorithms are sensitive to the size of a feature (larger features are more important)
- If we don't know how important different features are then it makes sense to normalise our features

Normalising Features

- Measuring a feature in millimeters or kilometers is going to make a lot of difference to the size of that feature
- Many learning algorithms are sensitive to the size of a feature (larger features are more important)
- If we don't know how important different features are then it makes sense to normalise our features

Normalising Features

- Measuring a feature in millimeters or kilometers is going to make a lot of difference to the size of that feature
- Many learning algorithms are sensitive to the size of a feature (larger features are more important)
- If we don't know how important different features are then it makes sense to normalise our features, E.g.

$$x_i^\alpha \leftarrow \frac{x_i^\alpha - \hat{\mu}_i}{\hat{\sigma}_i}, \quad \hat{\mu} = \frac{1}{m} \sum_{\beta=1}^m x_i^\beta, \quad \hat{\sigma}_i^2 = \frac{1}{m-1} \sum_{\beta=1}^m (x_i^\beta - \hat{\mu}_i)^2$$

Outline

1. Over-fitting?
2. Controlling Complexity
3. Hidden structure
4. **Regularisation**



Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2$$

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

(Good to normalise features)

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

(Good to normalise features)

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

(Good to normalise features)

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

(Good to normalise features)

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x} = \sum_{i=1}^p w_i x_i$$

varies rapidly as we change x_i

Explicit Regularisation

- As you've seen in the foundations of ML course, we can modify our error function to choose smoother functions

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \|\mathbf{w}\|^2$$

(Good to normalise features)

- Second term is minimised when $w_i = 0$
- If w_i is large then

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{w}^\top \mathbf{x} = \sum_{i=1}^p w_i x_i$$

varies rapidly as we change x_i

Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^T x_k - y_k)^2$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^T x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

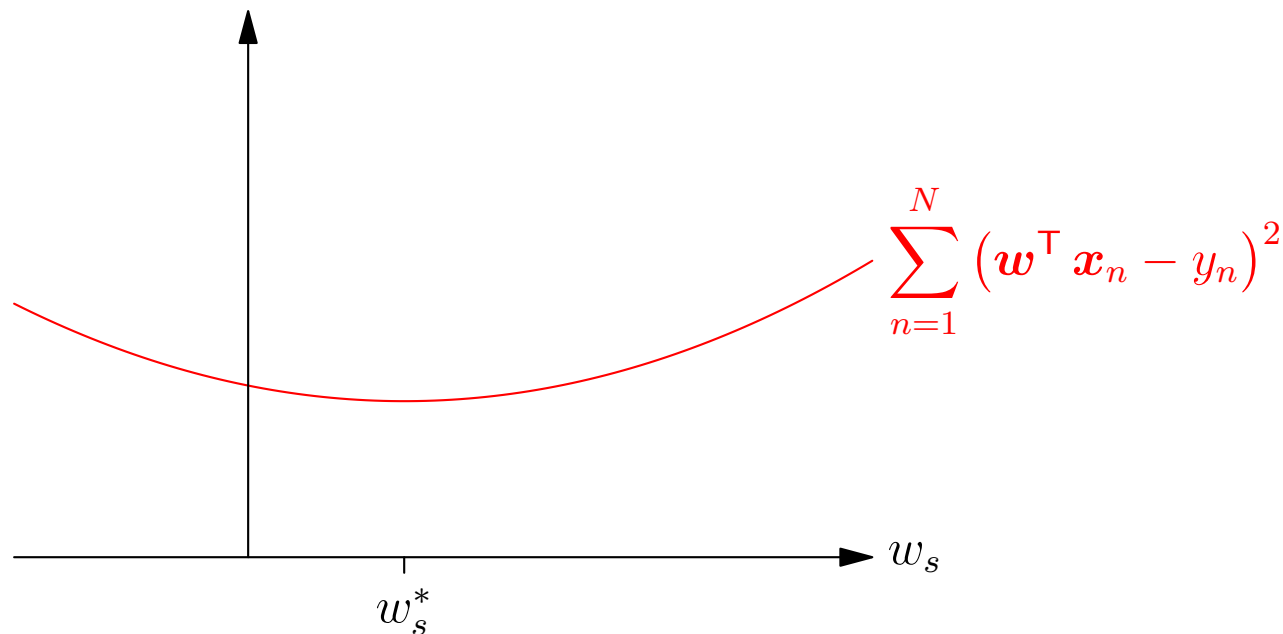
- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

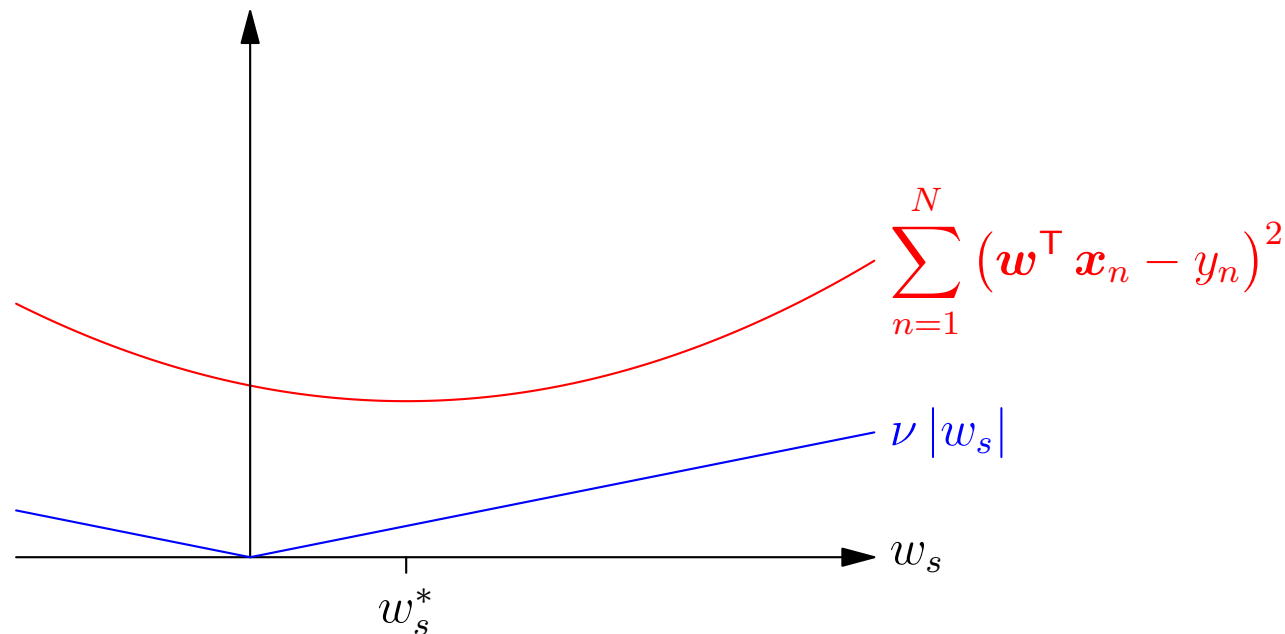


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

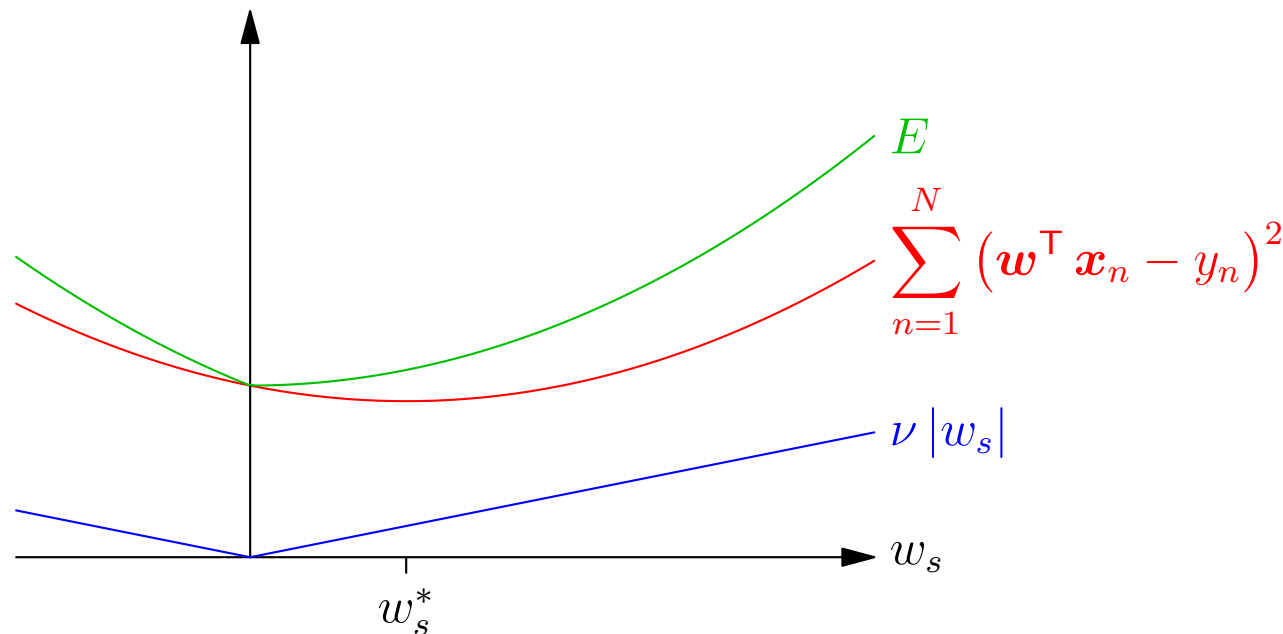


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^T x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

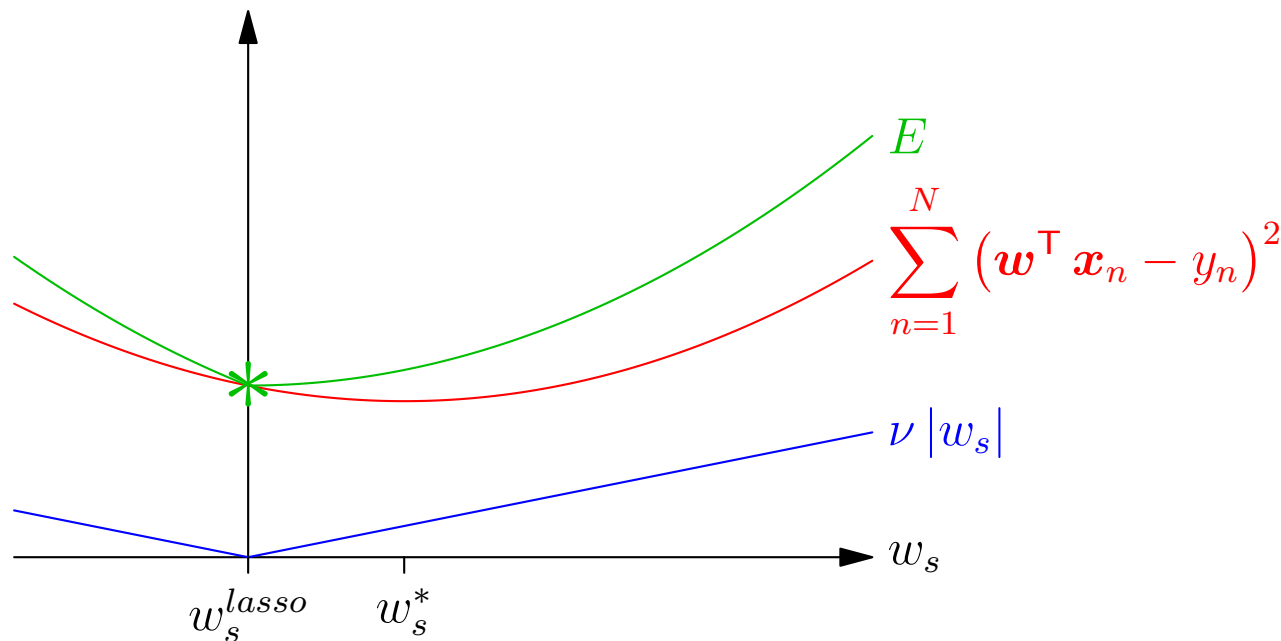


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

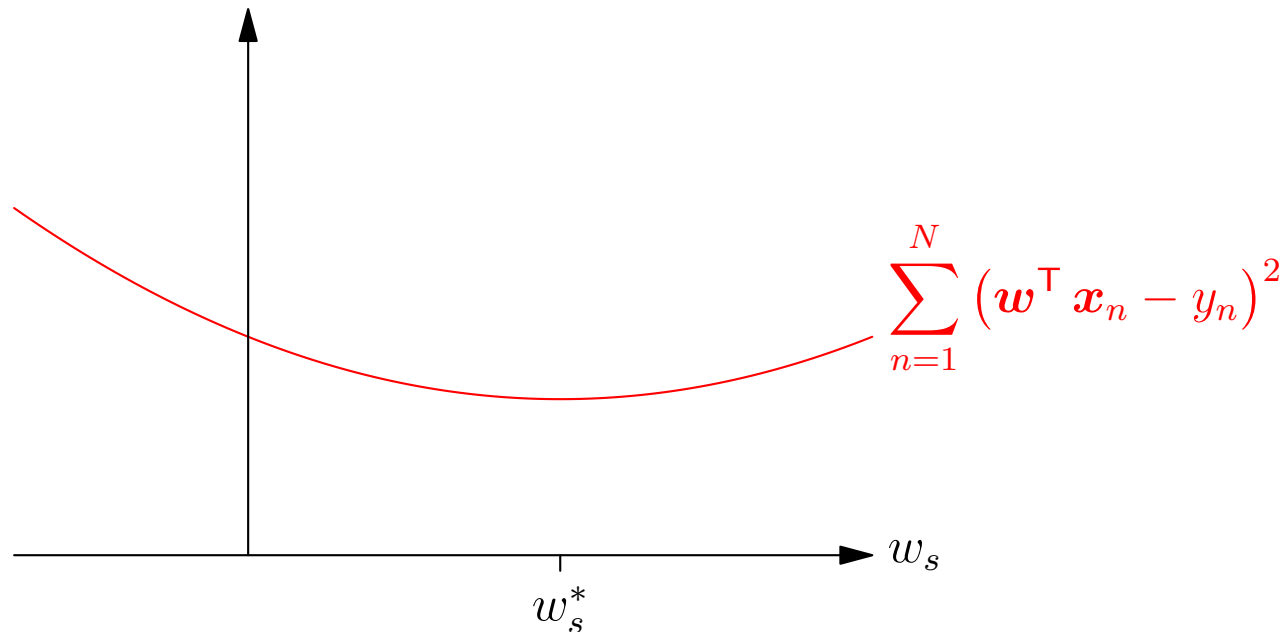


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^T x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

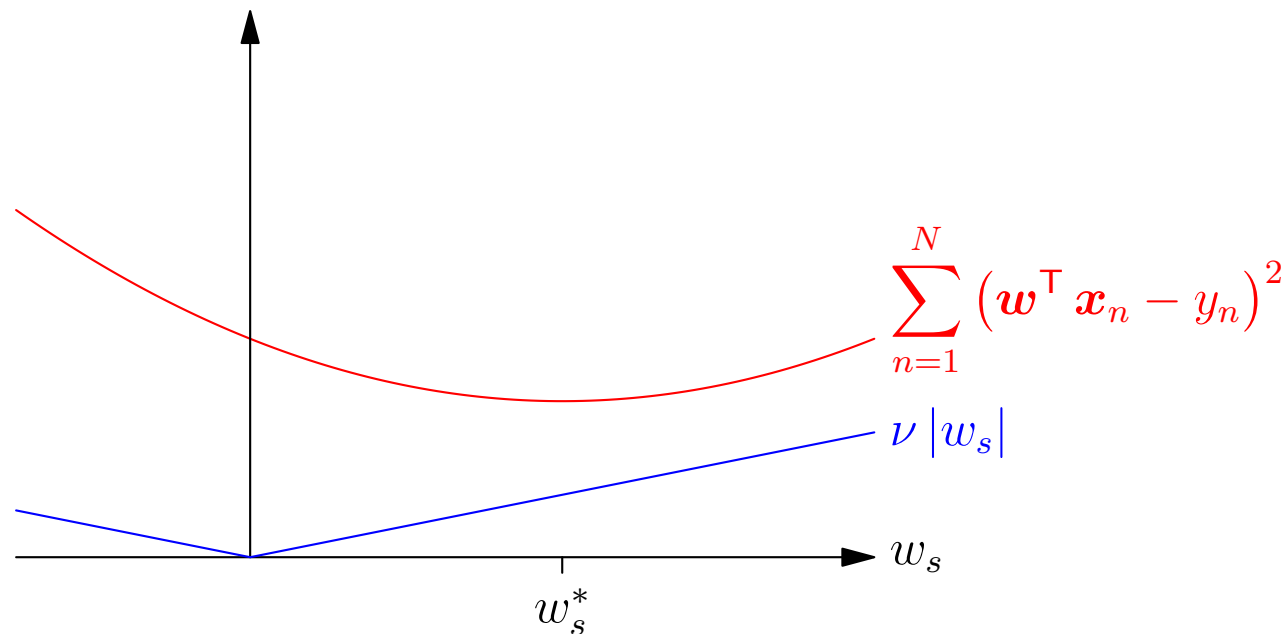


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^T x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

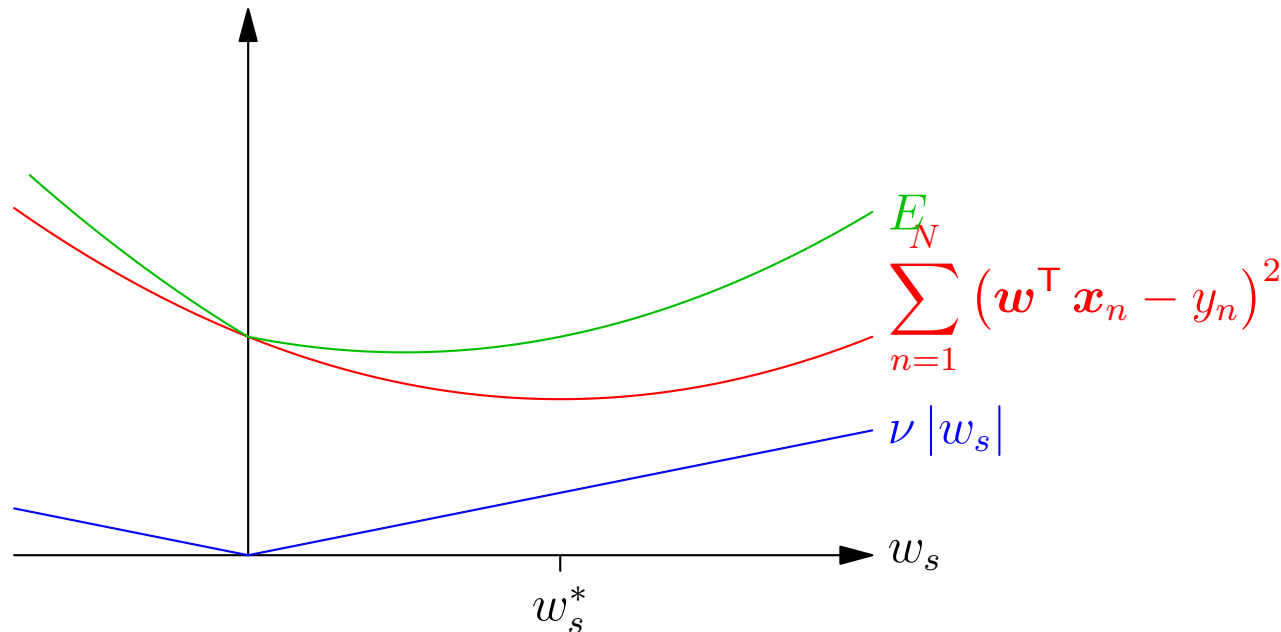


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^\top x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

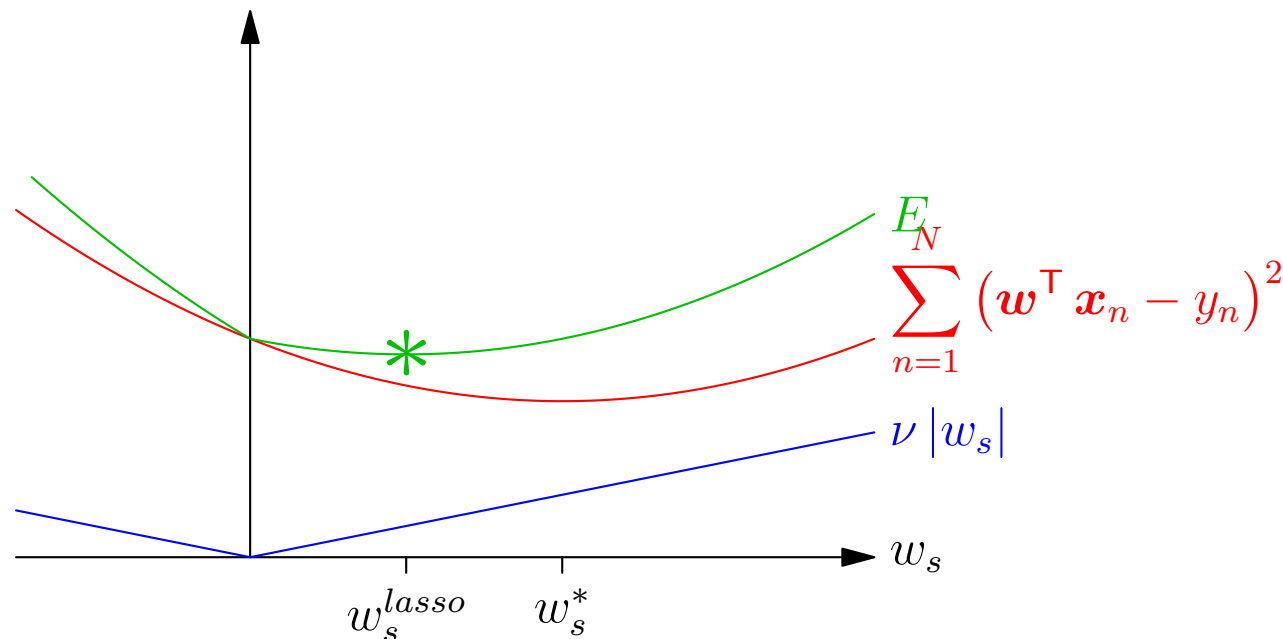


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^\top x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

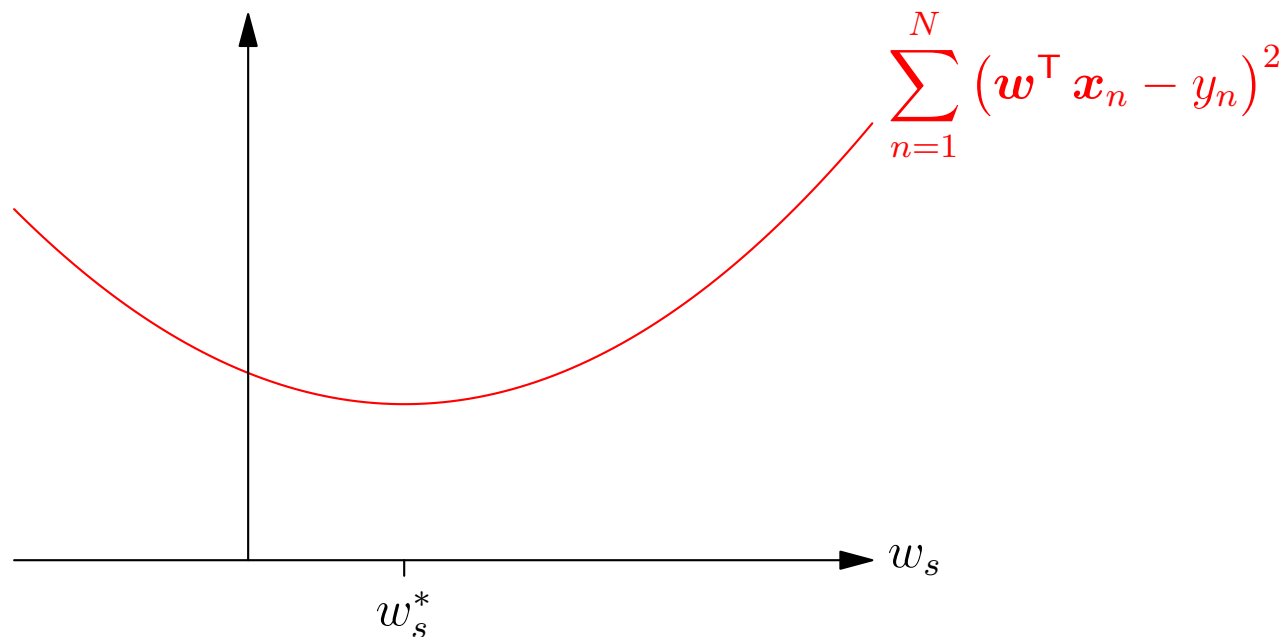


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

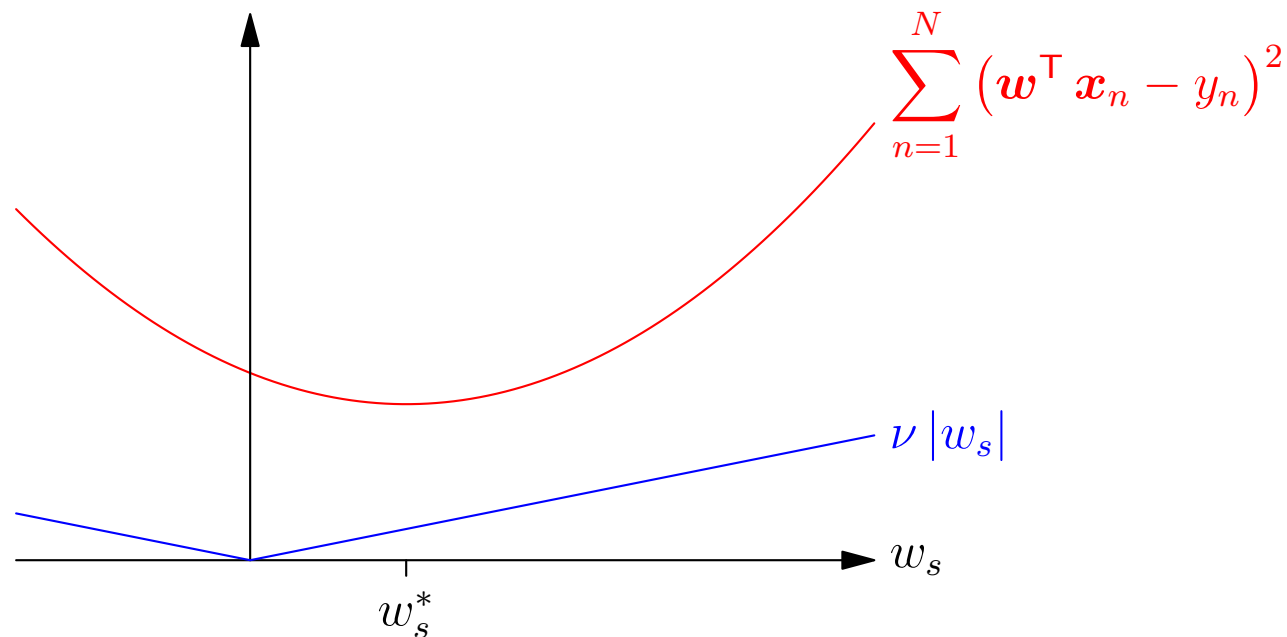


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

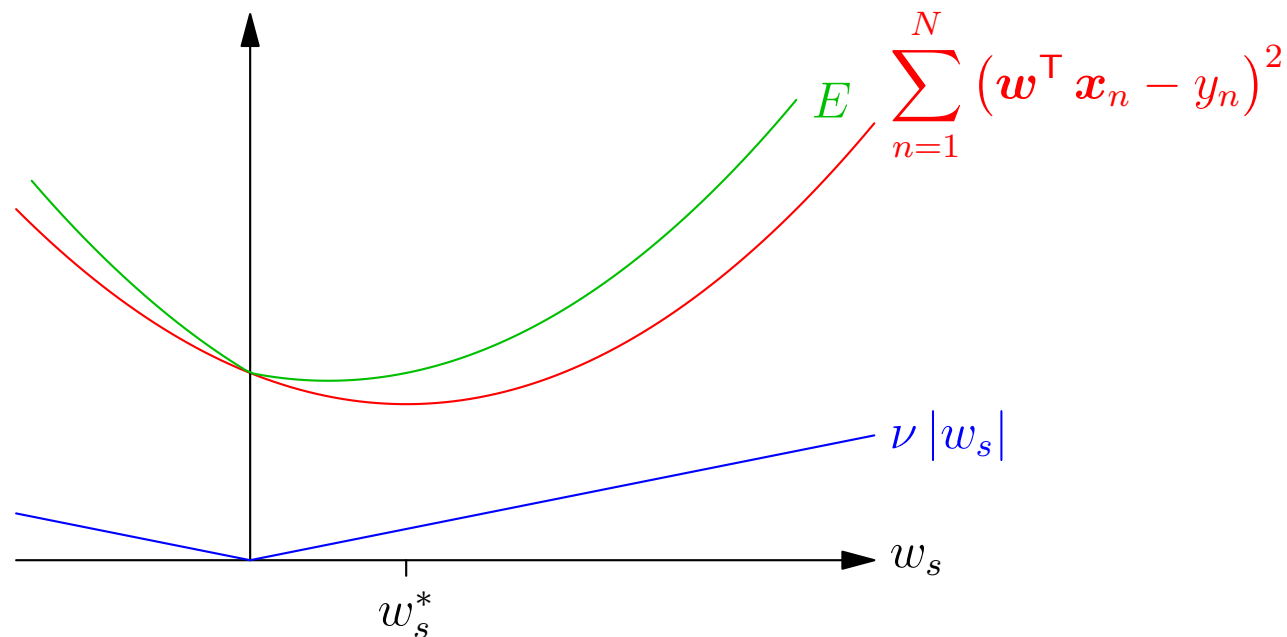


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error

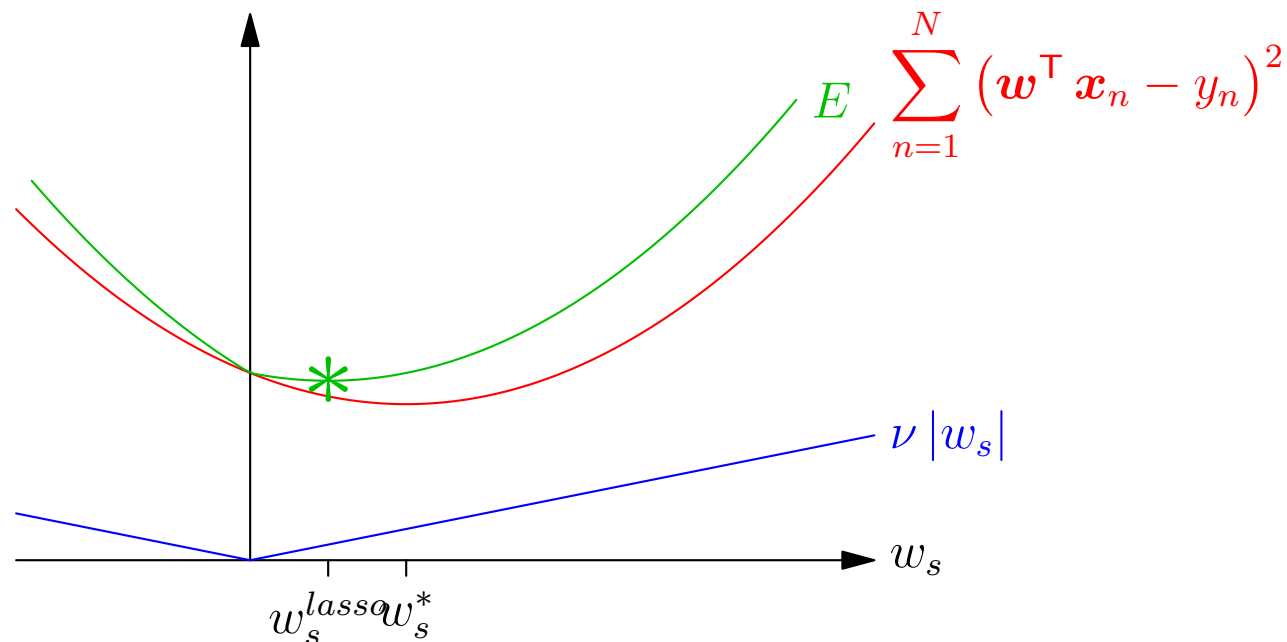


Lasso

- We can use other regularisers

$$L = \sum_{k=1}^m (w^\top x_k - y_k)^2 + \nu \sum_{i=1}^p |w_i|$$

- Spurious features (e.g. colour of flag on energy consumption) will give us a small improvement in training error



Implicit Regularisation

- In the last two examples we added an explicit regularisation term that made the function we learnt simpler
- Some learning machines do this less explicitly
- Some deep learning architectures do subtle averaging
- Sometimes the architecture biases the machine to find a simple solution

Implicit Regularisation

- In the last two examples we added an explicit regularisation term that made the function we learnt simpler
- Some learning machines do this less explicitly
- Some deep learning architectures do subtle averaging
- Sometimes the architecture biases the machine to find a simple solution

Implicit Regularisation

- In the last two examples we added an explicit regularisation term that made the function we learnt simpler
- Some learning machines do this less explicitly
- Some deep learning architectures do subtle averaging
- Sometimes the architecture biases the machine to find a simple solution

Implicit Regularisation

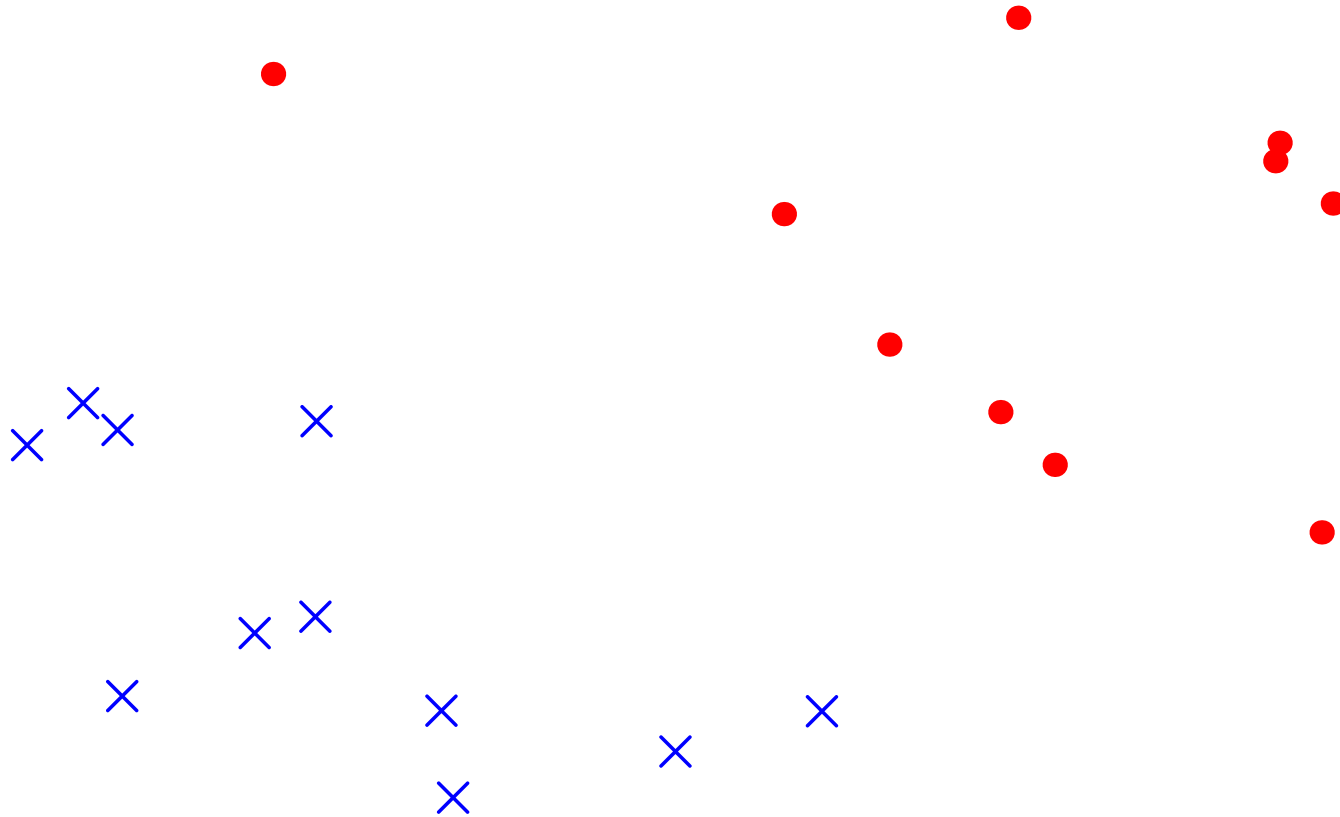
- In the last two examples we added an explicit regularisation term that made the function we learnt simpler
- Some learning machines do this less explicitly
- Some deep learning architectures do subtle averaging
- Sometimes the architecture biases the machine to find a simple solution

Maximum Margin Machines

- Perceptrons have many options to separate data

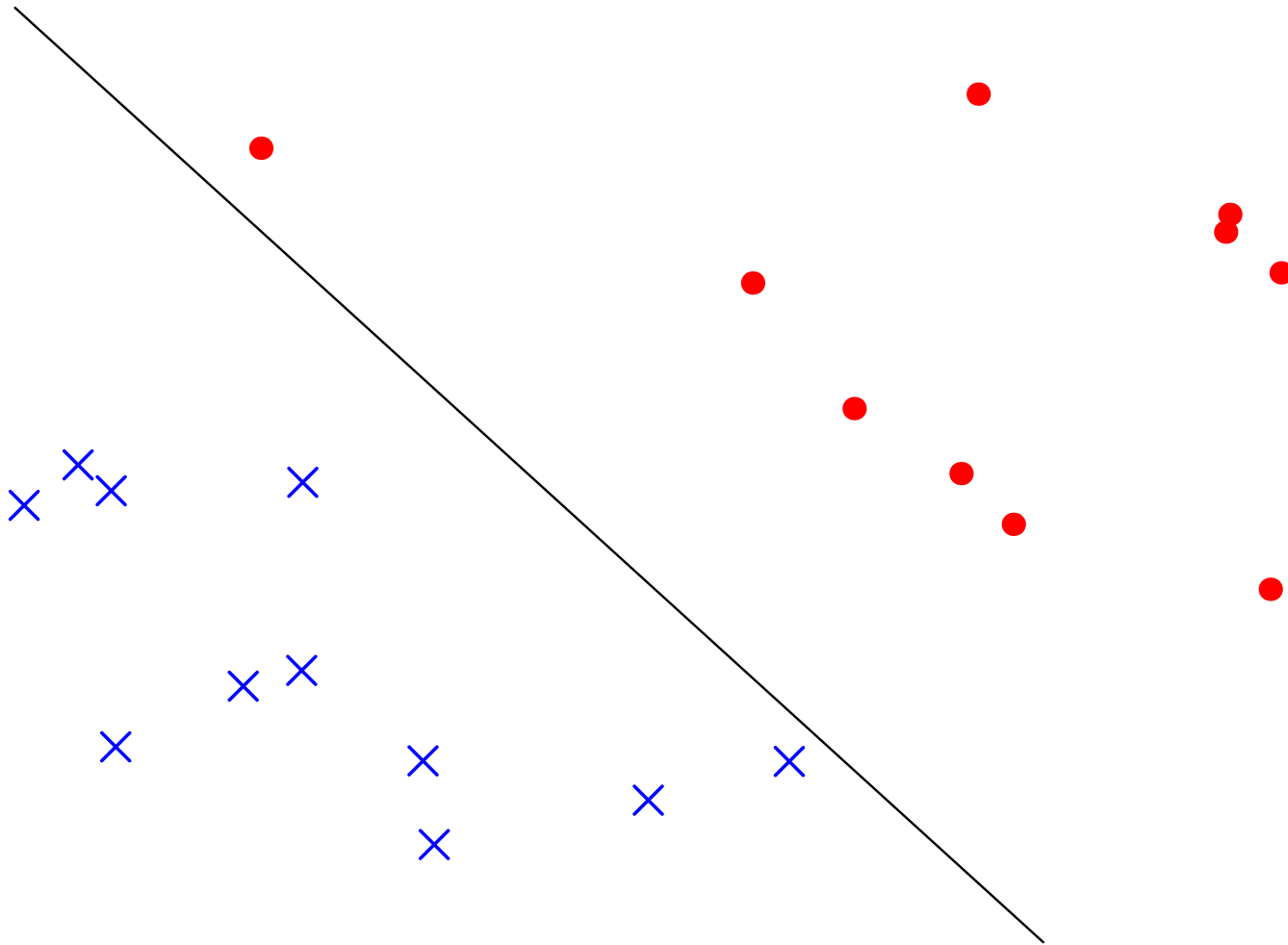
Maximum Margin Machines

- Perceptrons have many options to separate data



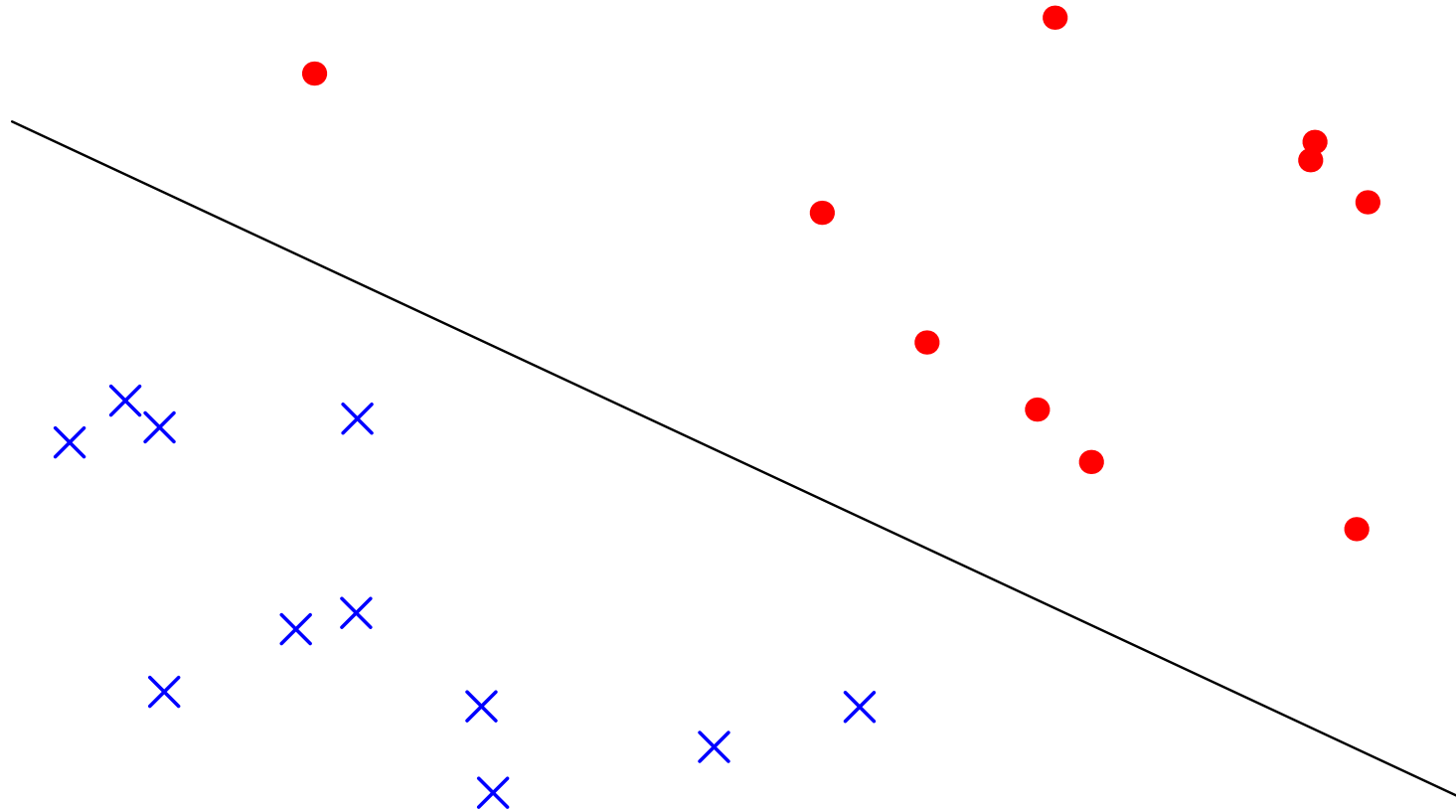
Maximum Margin Machines

- Perceptrons have many options to separate data



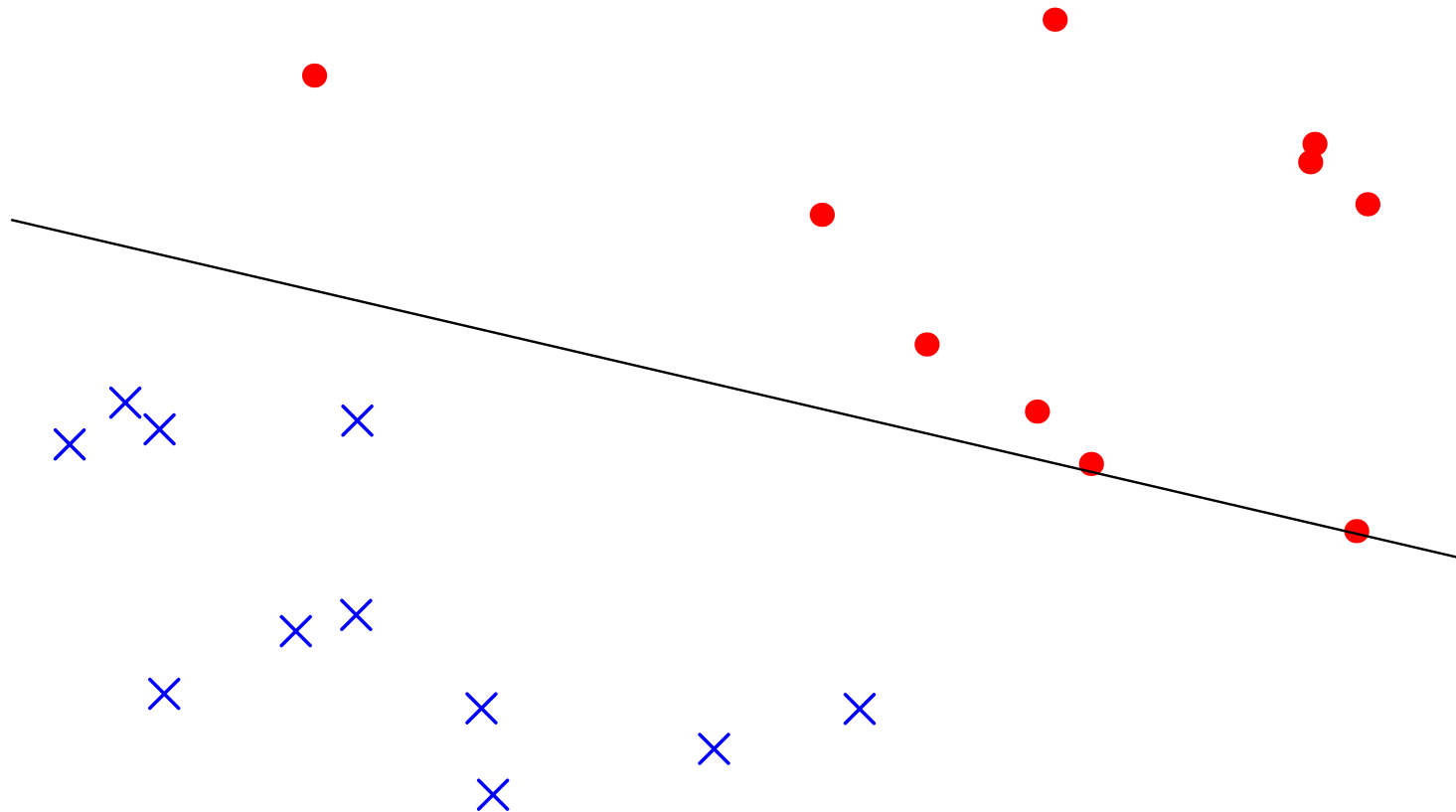
Maximum Margin Machines

- Perceptrons have many options to separate data



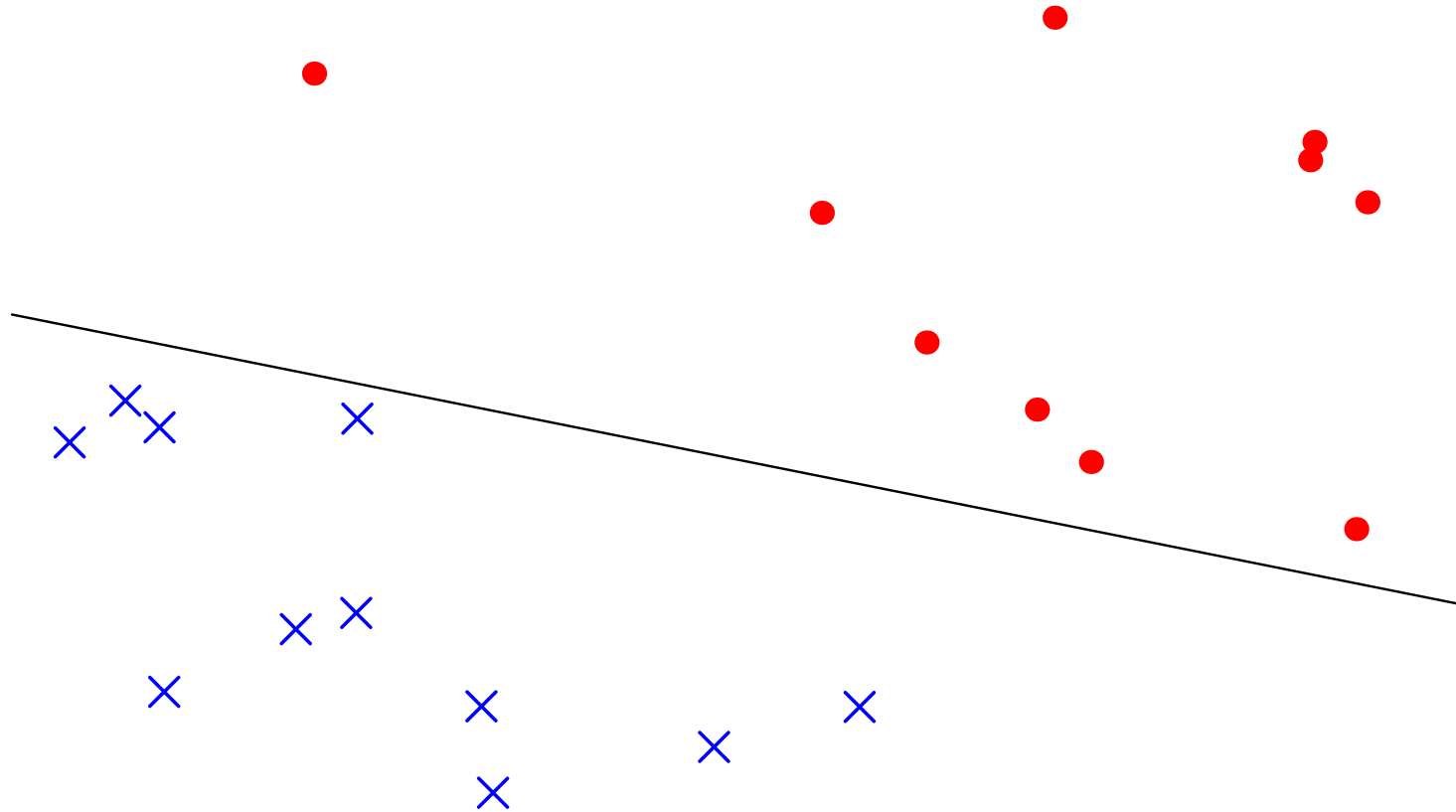
Maximum Margin Machines

- Perceptrons have many options to separate data



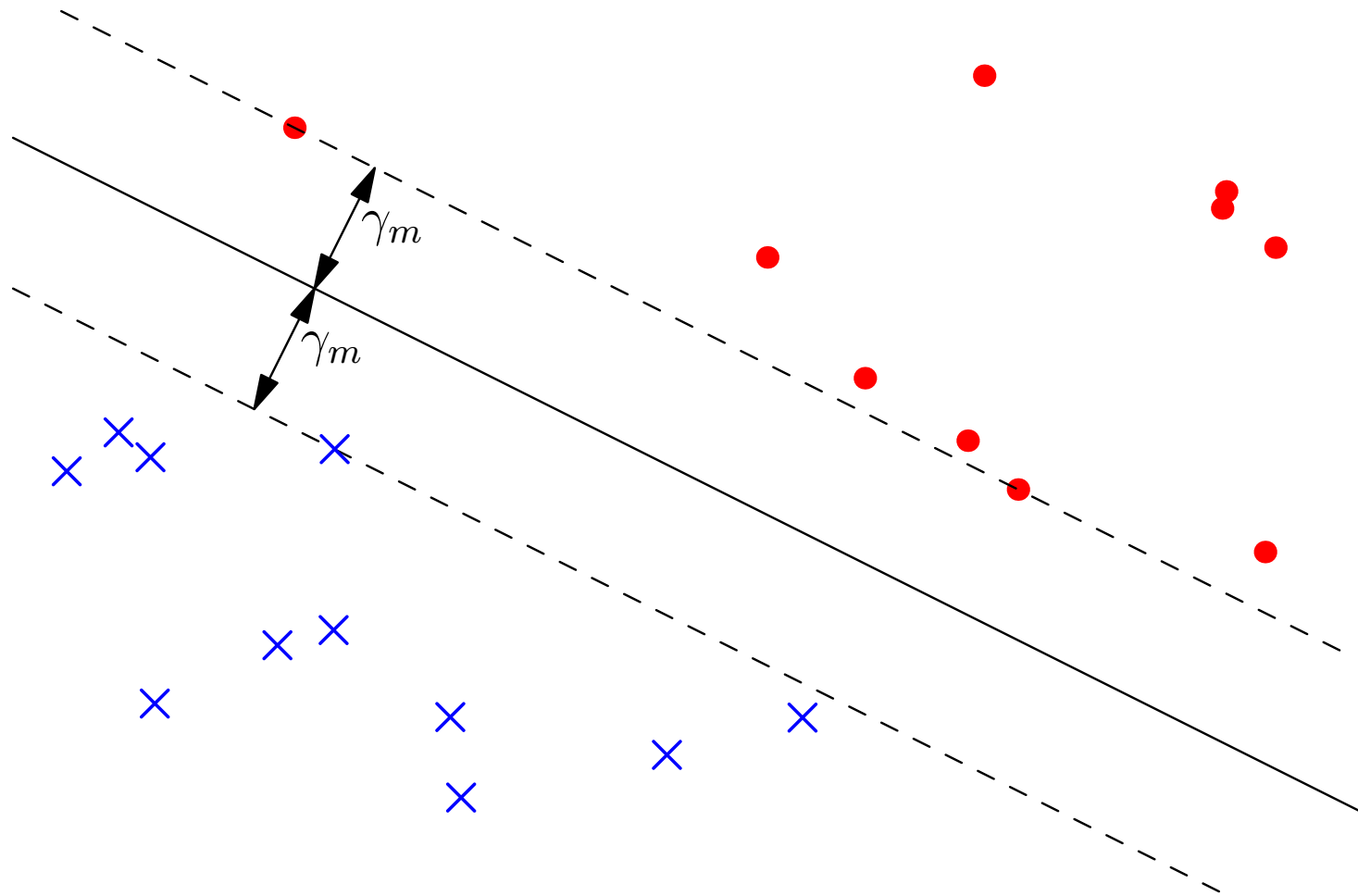
Maximum Margin Machines

- Perceptrons have many options to separate data



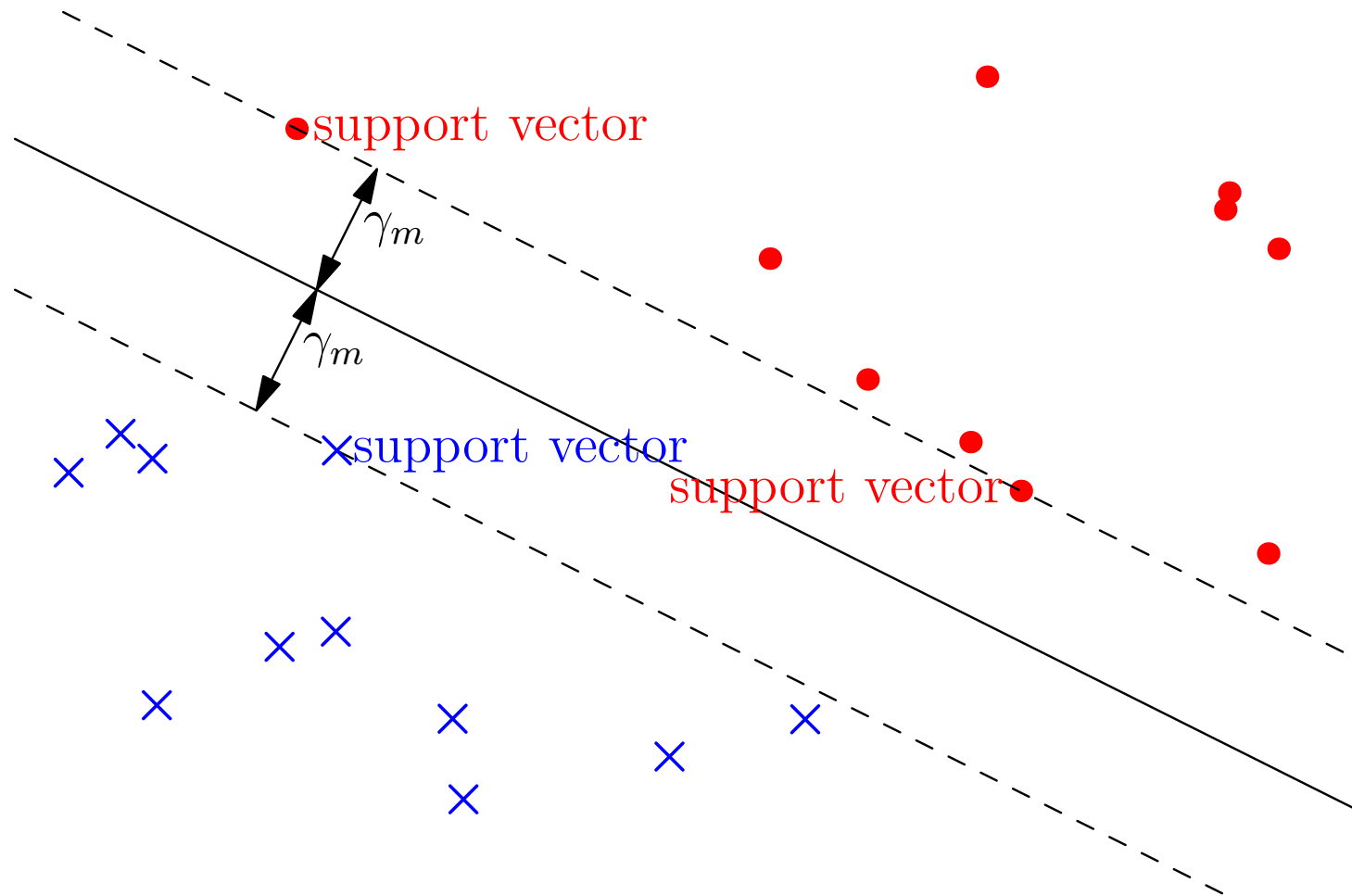
Maximum Margin Machines

- Perceptrons have many options to separate data



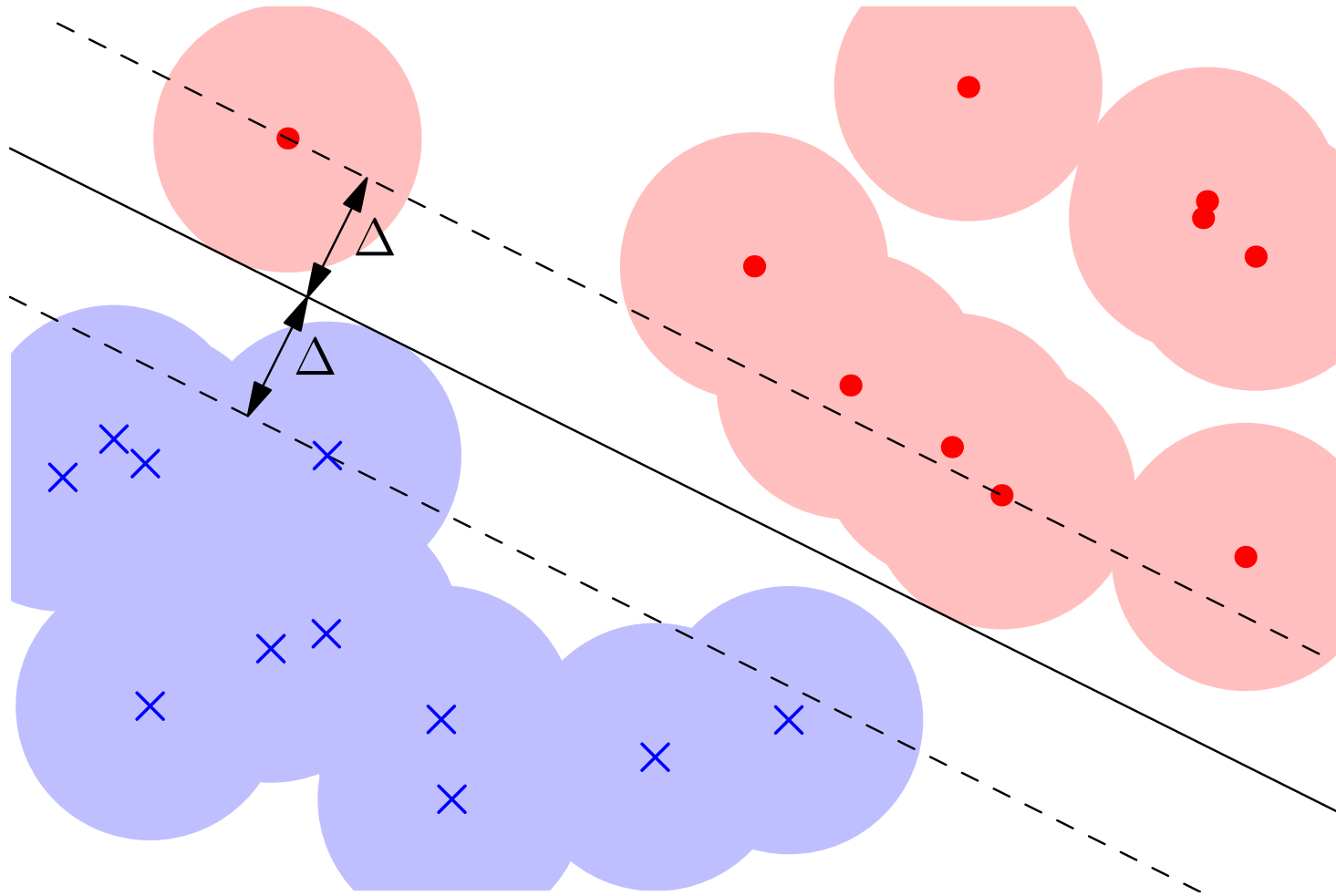
Maximum Margin Machines

- Perceptrons have many options to separate data



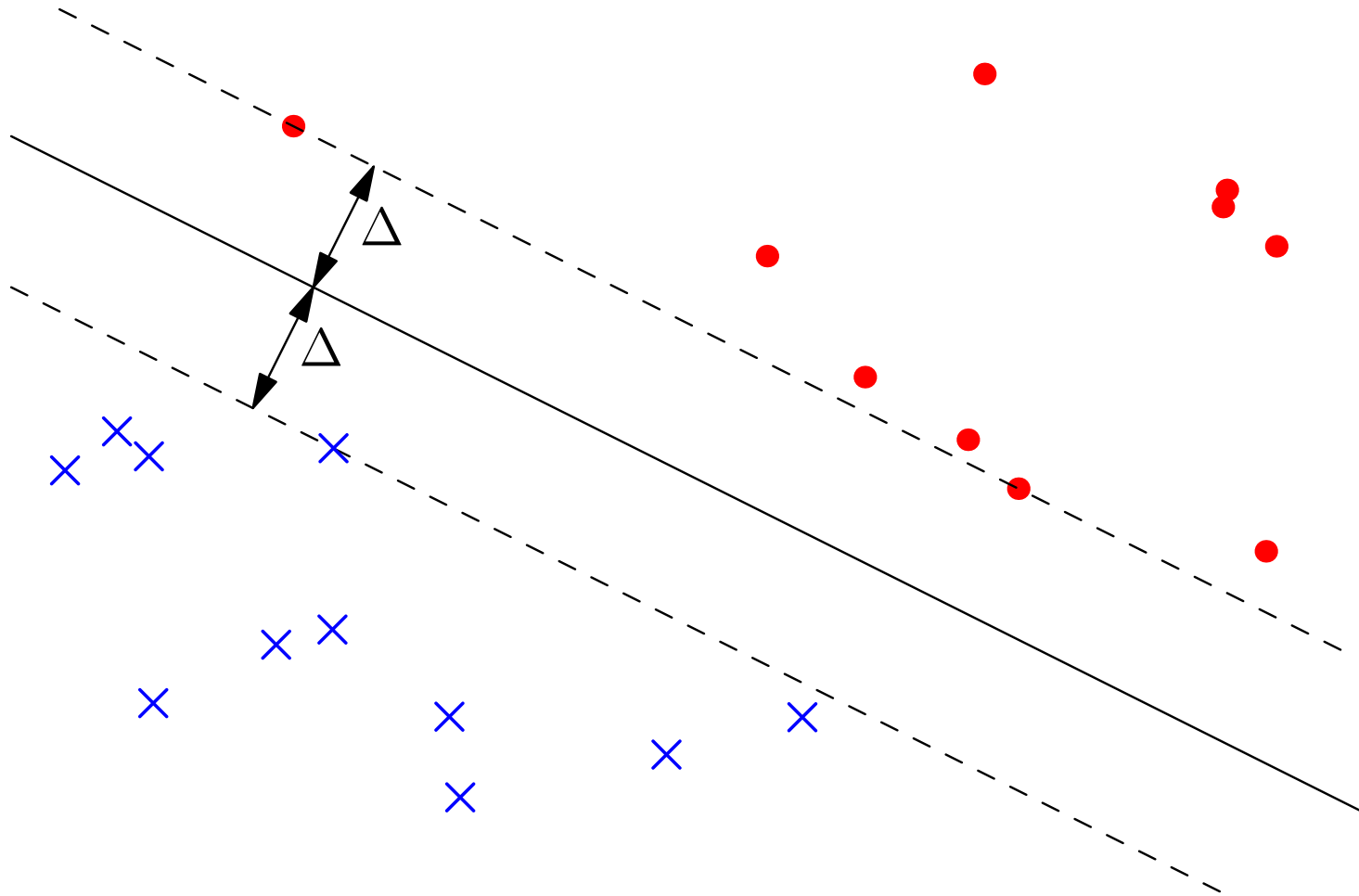
Maximum Margin Machines

- Perceptrons have many options to separate data



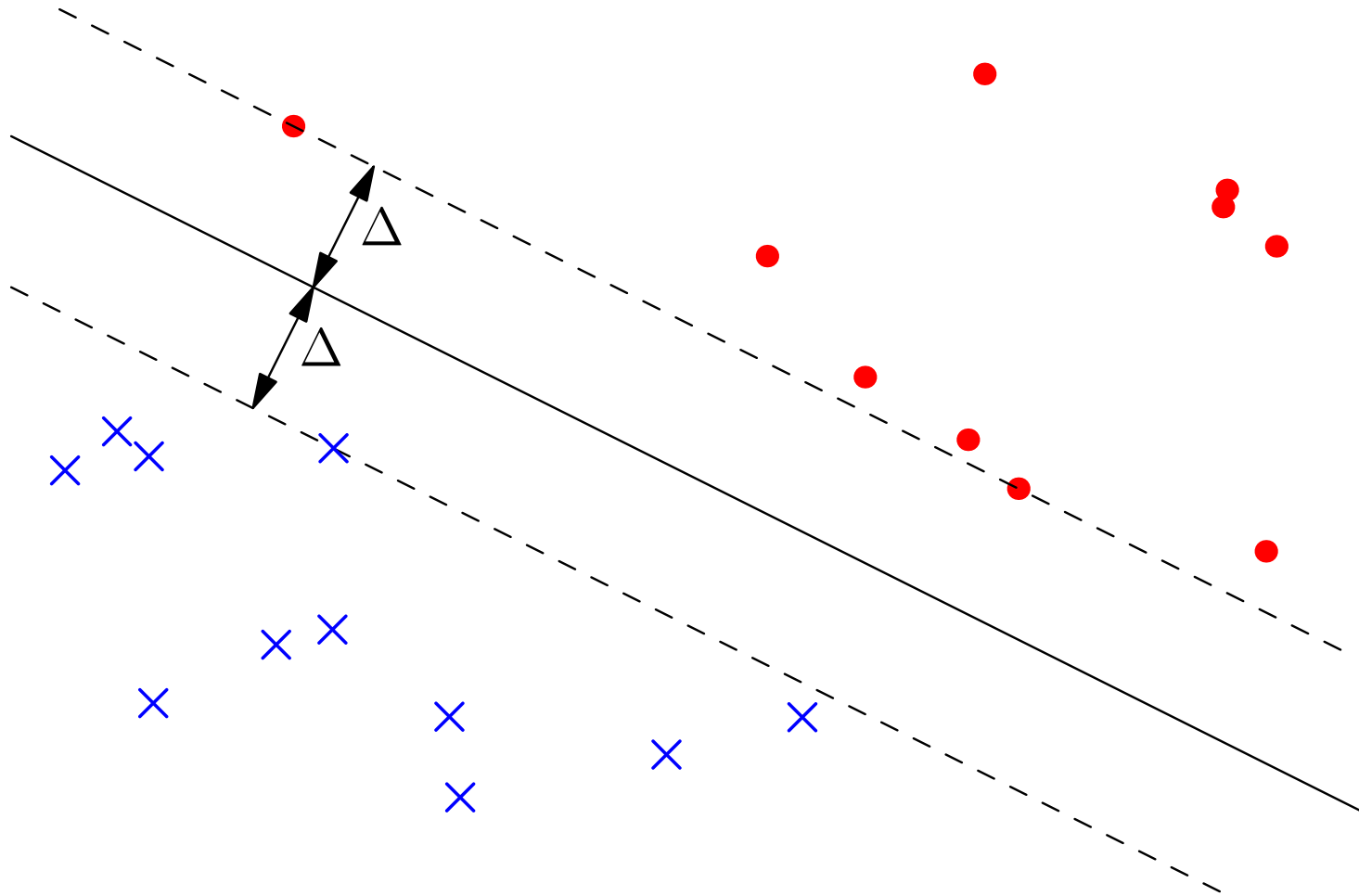
Maximum Margin Machines

- Perceptrons have many options to separate data



Maximum Margin Machines

- Perceptrons have many options to separate data



- SVMs choose the machine with the biggest margins

Success of SVMs

- SVMs regularise themselves by choosing the machine with the largest margin
- This ensures maximum stability to noise on the data
- It leads to very good generalisation on small datasets—usually beats everything else

Success of SVMs

- SVMs regularise themselves by choosing the machine with the largest margin
- This ensures maximum stability to noise on the data
- It leads to very good generalisation on small datasets—usually beats everything else

Success of SVMs

- SVMs regularise themselves by choosing the machine with the largest margin
- This ensures maximum stability to noise on the data
- It leads to very good generalisation on small datasets—usually beats everything else

Success of SVMs

- SVMs regularise themselves by choosing the machine with the largest margin
- This ensures maximum stability to noise on the data
- It leads to very good generalisation on small datasets—usually beats everything else

Success of SVMs

- SVMs regularise themselves by choosing the machine with the largest margin
- This ensures maximum stability to noise on the data
- It leads to very good generalisation on small datasets—usually beats everything else
- But you still need to normalise the features

Success of SVMs

- SVMs regularise themselves by choosing the machine with the largest margin
- This ensures maximum stability to noise on the data
- It leads to very good generalisation on small datasets—usually beats everything else
- But you still need to normalise the features
- You also need to tune its hyper-parameters (C and sometimes γ)

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
- Most machines have hyper-parameter that you tune to match the machine to the data
- Really clever machines try to do this matching automatically

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
- Most machines have hyper-parameter that you tune to match the machine to the data
- Really clever machines try to do this matching automatically

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
- Most machines have hyper-parameter that you tune to match the machine to the data
- Really clever machines try to do this matching automatically

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
- Most machines have hyper-parameter that you tune to match the machine to the data
- Really clever machines try to do this matching automatically

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
- Most machines have hyper-parameter that you tune to match the machine to the data
- Really clever machines try to do this matching automatically

Lessons

- Machine learning isn't magic
- It works when the learning machine is well attuned to the problem
- Sometimes you can help by preprocessing your data
- Sometimes there is a regularisation term that helps select a simpler machine
- Most machines have hyper-parameter that you tune to match the machine to the data
- Really clever machines try to do this matching automatically